



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)

Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade



SCHOOL OF ELECTRONICS, ELECTRICAL AND BIOMEDICAL TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PRESENTS

ELECTROCOMM



VOLUME 15
ISSUE 30



ABOUT THE DEPARTMENT

The Department of Electronics and Communication Engineering (ECE) at Kalasalingam Academy of Research and Education (KARE) is pioneer in academic excellence and innovative research. Accredited by ABET (Accreditation Board for Engineering and Technology, USA) and NBA (National Board of Accreditation, India), the department adheres to global standards in education and research.

The department has been strengthened by support from prestigious initiatives such as:

- DST-FIST (Department of Science and Technology - Fund for Improvement of S&T Infrastructure), enhancing state-of-the-art laboratories and fostering advanced research.
- AICTE MODROB funding, which has facilitated the modernization of the Embedded Systems and Robotics Laboratory, equipping it with cutting-edge tools for innovation in microcontrollers and robotics.
- Through collaborations with industry partners, the department offers industry-supported courses designed to align academic learning with real-world applications. Notable partnerships include Nanochip Technologies, P2F Semi, and MathWorks, enabling students to gain expertise in VLSI, IoT, embedded systems, and sustainable technologies relevant to industrial demands.
- The department's focus on experiential learning includes hands-on projects, industry internships, and problem-based learning, ensuring graduates are well-equipped to address global engineering challenges and contribute to sustainable development.

Department of Electronics and Communication Engineering

INSTITUTE VISION:

To be a University of Excellence of International Repute in Education and Research.



INSTITUTE MISSION:



To provide a scholarly teaching-learning ambience which results in creating graduates equipped with skills and acumen to solve real-life problems.
To promote research and create knowledge for human welfare, rural and societal development.
To nurture entrepreneurial ambition, industrial and societal connect by creating an environment through which innovators and leaders emerge.

DEPARTMENT VISION:

To strive towards excellence in Electronics and Communication engineering through teaching, experiential learning, quality research and scholarship, while adhering to ethical and societal requirements



DEPARTMENT MISSION:

To help students achieve their goals by recognizing, identifying, and fostering their unique strengths through quality education and cutting-edge research training.
To imbibe ability in the students to solve real life problems as per need of the society through nurturing their skills, creative thinking, and research acumen.
To create an elite workforce, without compromise in the ethics and societal values.

Program Educational Objectives (PEOs) (After June 2021)

PEO1: Technical Proficiency

Succeed as creative, productive, and valued engineers in their career by applying the technical knowledge and skills gained.

PEO2: Professional Growth

Continue to develop professionally through life-long learning, higher education, and research expertise to meet the expectations of their organisation and the society.

PEO3: Management Skills

Exhibit the management qualities in a responsive, ethical, and innovative manner.

B.TECH ECE PROGRAMME SPECIFIC OUTCOMES

At the end of the programme, the students will be able to:

PSO1: Knowledge:

Apply the knowledge of engineering and basic sciences to design, develop, test components and systems for applications including Signal Processing, VLSI Design, Embedded, Communication Engineering and Networking

PSO2: Skills:

Solve the complex engineering problems, using latest techniques, tools, along with the needed skills with an understanding of societal, environmental, safety, legal and cultural impacts of the solution.

PSO3: Attitude:

Apply the contextual knowledge of engineering to function effectively as an individual or a leader in multidisciplinary environments

B.TECH ECE PROGRAMME OUTCOMES (R2018)

At the end of the programme, the students will be able to:

PO1: Engineering knowledge:

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialisation to the solution of complex engineering problems.

PO2: Problem analysis:

Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions:

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems:

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage:

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society:

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

B.TECH ECE PROGRAMME OUTCOMES (R2018)

PO7: Environment and sustainability:

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics:

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work:

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication:

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance:

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning:

Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

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Dear Readers

Welcome to the latest issue of ELECTROCOMM -Volume 15, Issue 30. It is with great pleasure and enthusiasm that we present to you a diverse collection of stories, insights, and inspiration curated for your reading pleasure.

In the fast-paced world we live in ELECTROCOMM aims to be your guide, offering a moment of respite and discovery.

As we flip through these pages together, you will encounter thought-provoking articles, captivating visuals, and interviews with individuals who are making waves in their respective fields. Our goal is to create a space for meaningful dialogue, celebration of accomplishments, and exploration of the ideas shaping our world.

A magazine is more than just a collection of words and images; it's a journey. It's an exploration of new concepts, a celebration of diversity, and a platform for voices that deserve to be heard.

We extend our deepest gratitude to our talented contributors, whose creativity and dedication have brought this magazine to life. We also want to thank you, our readers, for your continued support and for being an essential part of the ELECTROCOMM community.

As you delve into the pages ahead, we encourage you to immerse yourself fully in the stories, connect with the ideas, and share your thoughts with us. Your feedback is invaluable and helps us grow.

Dr. A. Muthukumar
HOD, ECE



Dear Readers

Warm Greetings to all!!!

It gives me immense joy and satisfaction to pen in the 29th Issue, of our Department Magazine 'ELECTROCOMM' on this occasion as Editor. Magazines are the perfect platform that develops one's creativity. The biggest blocks to creativity is the inability to allow a thought or insight to have the space to become a possibility. Some of the inspired idea are the ones that start as crazy concepts. By allowing them to possibility, they may blossom into something usable. The Magazine aims to encourage creativity among young minds and inspire self-motivation. This issue includes articles, poems, sketches, photographs and technical, non- technical writeups and stories.

In each one, the contributors demonstrated their best aspects of their personalities through self-expression.

I would like to thank each and every contributor for their active participation in making the Magazine to float in our hands.

As an Editor, I hope you too enjoy reading this Magazine as much as we have enjoyed creating it.

A handwritten signature in black ink, appearing to read 'Abha'.

Dr. Abha Dargar
Asst. Prof, ECE

MEET THE TEAM

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Droupadi
Murmu

Droupadi Murmu

Droupadi Murmu

Draupadi Murmu was the governor of Soren's Jharkhand. She was descended from the Santhal tribals of Odisha. The BJP-led National Democratic Alliance (NDA) has selected Murmu as its presidential candidate for the election that is scheduled for next month. Both Odisha and Jharkhand are states rich in tribal culture. The NDA and RSS view Murmu's candidature as an integral component of Prime Minister Narendra Modi's larger dharmic project, not just a token gesture as the BJP's fierce critics claim. This is because they see it as a well-thought-out plan to dramatically expand the reach of an inclusive, compassionate Hinduism among the tribal population in Jharkhand and throughout India. The goal of this strategy is not to place marginalized communities on an equal footing with mainstream Indian society, but rather to stretch the boundaries of Hinduism to the maximum extent possible. With strong political and even more important social and religious overtones, it seeks to dismantle the prejudiced tribal vs non-tribal dichotomy that Indian politicians and colonial authorities built over the last 150 years. The goal is to abolish the deeply splintered identity politics that the Nehruvian state normalized and legitimized, with the help of ideologically driven scholarship that aimed to deepen rifts within Hindu society.

Personal Life:

On June 20, 1958, in Uparbeda village in the Baidaposi area of Rairangpur, Odisha, Droupadi Murmu was born into a Santali family. Biranchi Narayan Tudu, her father, was a farmer. Her grandpa and father served as the customary Sarpanch, or chairman, of the Gram Panchayat, or village council. Her paternal grandmother, Durgi Tudu, was the name her family gave her; Batti, her previous monicker, was changed to Puti. Her teacher at school gave her the new name Droupadi, and she had previously had her name altered to Durpadi and Dorpdi.



Murmu attended the Upar Beda local primary school to study elementary education. She relocated to Bhubaneswar at age five to further her education. She completed Girl's High School Unit-2 for her secondary education, and Rama Devi Women's College awarded her a B.A.

She had two boys and a daughter with financier Shyam Charan Murmu (1 April 1958 – 1 August 2014) whom she married in 1980. In the seven years between 2009 and 2015, her husband, two kids (Lexman died in suspicious circumstances, and Sipun died in a car accident), mother, and brother all passed away. She belongs to the Brahma Kumaris spiritual group. Jawaharlal Nehru, the first prime minister of India, is admired by Murmu. She has also praised B. R. Ambedkar and Mahatma Gandhi.

Political Career:

As an independent candidate running for a women's reserved seat in the Rairangpur Nagar Panchayat, Droupadi Murmu was chosen to serve as a councillor in 1997. After that, she joined the BJP, or Bharatiya Janata Party.

She was elected to the Odisha Legislative Assembly in 2000 from the Rairangpur Assembly constituency, and she held office for two terms, from 2000 to 2009. She served as the Minister of State with Independent Charge for Commerce and Transportation during the BJP and BJD coalition administration in Odisha from March 6, 2000, to August 6, 2002, and for the development of Fisheries and Animal Resources from August 6, 2002, to May 16, 2004.

She was awarded the 2007 Nilkanth Award as the Odisha Legislative Assembly's Best MLA. As the BJD and BJP partnership came to an end, she lost the 2009 Lok Sabha election from the Mayurbhanj Lok Sabha constituency. She served as the district president of the BJP National Executive (ST Morcha) till her election in 2013.

Pathalgadi movement:

In 2016-2017, the Raghuraj Das Ministry proposed changes to the Santhal Pargana Tenancy Act of 1949 and the Chhotanagpur Tenancy Act of 1908, protecting the rights of indigenous communities in their land. The amendments gave tribes the ability to consent to the government using their territory for commercial purposes and to lease their land to others. The planned law was strongly opposed by the tribal people, and protests were held during the Pathalgadi movement. The measure was withdrawn in August 2017. Droupadi Murmu is the 15th President of India and the first elected president from one of India's declared tribal communities. She introduced the "Pradhanmantri TB Mukht Bharat" program that aims to end tuberculosis in the nation. She made history as the first Indian president to open the ten-day state celebration known as "Mysuru Dasara." She participated in ceremonies for several development projects and visited her home state of Odisha for the first time in November 2022. In April 2023, Murmu became the third president in history to fly in a sortie fighter jet. She spent about thirty minutes in the air, crossing the Himalayan ranges as well as the lowlands of Tezpur and Brahmaputra. Afterward, she called it a "thrilling experience." In May 2023, the opposition abstained from the unveiling of the New Parliament House, part of the Central Vista Project, and insisted that Murmu lead the inauguration. Nevertheless, Murmu later said she was proud and extremely satisfied with the inauguration and that it symbolized parliament's trust.



Sunil Chhetri

Sunil Chhetri, born on August 3, 1984, in Secunderabad, India, is a football icon whose prowess on the field has earned him admiration and respect not only in India but also on the global stage. As the captain of the Indian national football team and a prominent figure in Indian football history, Chhetri's journey is a testament to determination, skill, and passion for the beautiful game.

Chhetri's passion for football ignited at a young age, influenced by his father's love for the sport. His early years were spent honing his skills on the streets of his hometown and later representing his school and local clubs. His talent soon caught the eye of scouts, leading to his inclusion in the youth teams of Delhi-based clubs.



Chhetri's breakthrough came when he represented India at the U-20 Asian Championship in 2004, where he showcased his exceptional talent, catching the attention of football enthusiasts across the country. His performances earned him a call-up to the senior national team, marking the beginning of his illustrious international career.

Since his debut for the Indian national team in 2005, Chhetri has been instrumental in leading the team to numerous victories and milestones. He holds the record for the most international goals scored by an Indian player, surpassing football legends like Lionel Messi and Cristiano Ronaldo in terms of international goals scored.

In addition to his contributions to the national team, Chhetri has also enjoyed a successful club career both domestically and internationally. He has played for various clubs in India, including Bengaluru FC, where he has been a pivotal figure in the team's success, helping them win multiple titles in the Indian Super League (ISL) and the I-League. Off the field, Chhetri's influence extends beyond the realm of football. He is actively involved in philanthropic activities, using his platform to raise awareness about social issues and support charitable causes, earning him admiration as a role model and humanitarian.

Sunil Chhetri's legacy transcends mere statistics and accolades; it embodies the spirit of resilience, dedication, and excellence. As he continues to inspire generations of aspiring footballers and fans across the nation, his impact on Indian football and society at large is undeniable. Chhetri's journey serves as a beacon of hope, reminding us of the transformative power of sport to unite, inspire, and uplift communities.

Finally Sunil Chhetri's journey from the streets of Secunderabad to becoming the face of Indian football is a testament to his dedication, talent, and leadership. As he continues to inspire millions of aspiring footballers and fans across the nation, his legacy as a true footballing icon and ambassador for the sport is firmly cemented in the annals of Indian football history.



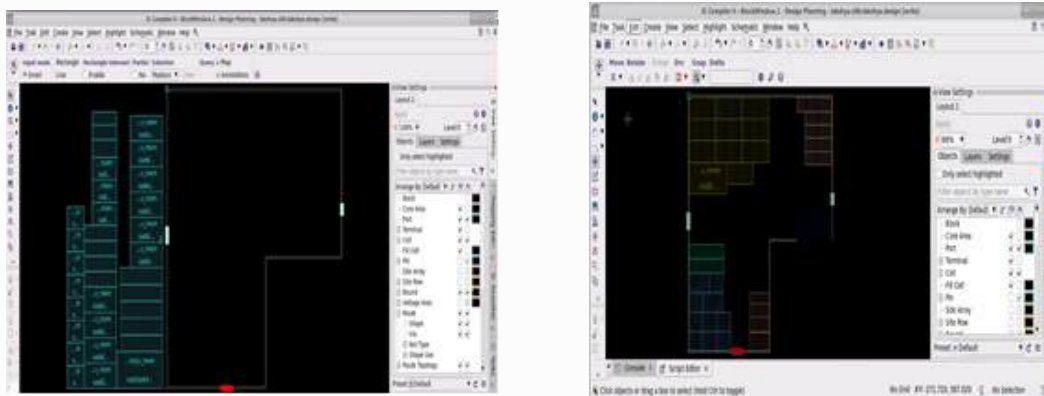


Faculty Articles

DESIGN OF PNR FLOW FOR BLOCK LEVEL CHIP TO OPTIMIZING LEAKAGE POWER

Background- To satisfy the future requirements of chip design, modifications must be made to software/hardware capabilities, design tools, and procedures. Engineers developed an efficient structured ASIC chip architecture to make these changes while concentrating on the design characteristics. ASIC design cycle that is mature and silicon-tested consists of several stages, including design conception, leakage power optimization, logical/physical implementation, design validation, and design verification.

Introduction: Technological advancements demand efficient semiconductor designs to enhance performance while minimizing power consumption. With the transition to nanometer-scale technologies, the design complexity increases, requiring robust methodologies. The PnR flow in Application-Specific Integrated Circuit (ASIC) design, including steps like floor planning, cell placement, and routing, is crucial for achieving optimal leakage power reduction. This study introduces a customized PnR flow, leveraging cutting-edge techniques to address challenges in leakage power optimization. It emphasizes hierarchical design flows, enabling early market delivery and improved scalability.



GUI of the floorplan (a) Pin, die, core area definition
(b) macro-placement in the central region.

The design flow begins with floor planning, defining the die and core area while arranging macros for efficient space utilization. Power plans use layered metal routing to ensure even power distribution, minimizing IR drops. During placement, high-density areas are addressed using congestion maps to improve layout efficiency. Clock tree synthesis reduces clock skew and power consumption through buffer optimization and low-power routing techniques. Finally, detailed routing optimizes signal integrity and minimizes parasitic effects, enhancing overall design efficiency. Pre- and post-leakage power analysis confirms the proposed methods' efficacy in reducing power dissipation across both static and dynamic parameters.

TABLE II PRE-LEAKAGE REPORT (A)
BRIEF REPORT (B) OVERALL REPORT

Cell Internal Power	1.06e+08 nW (82.8%)
Net Switching Power	2.21e+07 nW (17.2%)
Total Dynamic Power	1.29e+08 nW (100.0%)
Cell Leakage Power	6.29e+05 nW

TABLE II PRE-LEAKAGE REPORT (A)
BRIEF REPORT (B) OVERALL REPORT

Cell Internal Power	1.01e+08 nW
Net Switching Power	2.41e+07 nW
Total Dynamic Power	1.26e+08 nW
Cell Leakage Power	5.39e+05 nW

Power Group	Internal Power	Switching Power	Leakage Power	Total Power	(%)
Io_pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Memory	4.44e+07	2.59e+05	2.67e+05	4.49e+07	34.7%
Black_box	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Clock_network	5.59e+07	1.76e+07	2.59e+04	7.36e+07	57.0%
Register	4.15e+06	1.22e+06	1.54e+05	5.52e+06	4.3%
Sequential	4.49e+02	4.87e+02	3.02e+02	1.24e+03	0.0%
Combinationa	2.01e+06	3.00e+06	1.81e+05	5.19e+06	4.0%
Total	1.06e+08 nW	2.21e+0nW	6.2e+05nW	1.29e+8 nW	Total

The research paper is concluded by introducing some important ideas and factors for establishing block level PNR (Place and Route) flow of electronic equipment with a focus on increasing energy leakage. The main aim of this study is to achieve a balance between energy consumption and performance by integrating various methods and technologies in the design process. Using High-Voltage Threshold (HVT) cells becomes a key component in the effort to reduce leakage power. With their higher threshold voltages, these cells are very helpful in reducing leakage currents without sacrificing the overall performance of the device.



Dr. Shashi Kant Dargar,
Associate Professor, ECE

IONIZING RADIATION DEFECTS AND RELIABILITY OF GALLIUM NITRIDE-BASED III-V SEMICONDUCTOR DEVICES

Background: The Critical Challenge of Radiation in Electronics

In the unforgiving environments of space and high-energy applications, electronic devices face a relentless enemy: radiation. From the intense cosmic rays bombarding satellites to the harsh radiation in advanced defense systems, electronic components must withstand extreme conditions that would quickly destroy conventional electronics. Gallium Nitride (GaN) transistors have emerged as a promising solution, offering unprecedented resilience where traditional semiconductors falter.

The fundamental challenge lies in how radiation impacts electronic devices. When high-energy particles like protons, neutrons, and heavy ions collide with semiconductor materials, they can cause catastrophic damage. These particles can displace atoms in the crystal lattice, create trap states, and fundamentally alter the electrical characteristics of transistors. For critical applications in satellite communications, aerospace, and defense technologies, such degradation isn't just a technical problem—it's a mission-critical threat.

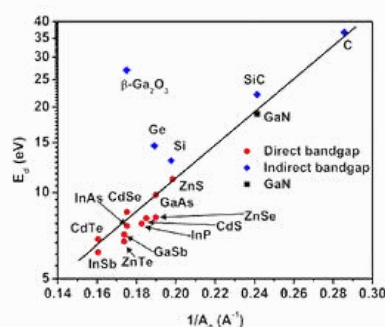


Fig. 1. Graphical representation of the relationship between threshold displacement energy and inverse lattice constant for several semiconductor crystals. GaN has a higher displacement energy than other crystals, displaying larger radiation hardness.

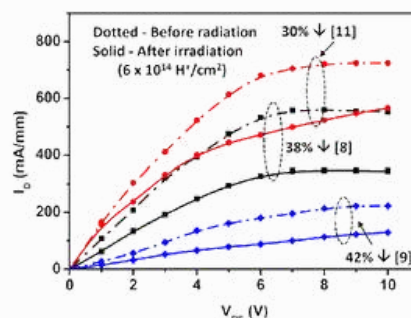


Fig. 2. Current-voltage characteristics derived from several works before and after irradiation.

Introduction: GaN - A Material Built for Extreme Conditions

Gallium Nitride (GaN) represents a breakthrough in semiconductor technology, particularly in its ability to withstand radiation. Unlike traditional semiconductor materials like silicon or gallium arsenide, GaN demonstrates remarkable structural integrity and electrical performance under extreme radiation conditions. Its higher lattice displacement energy means it can resist atomic displacement more effectively, providing a robust solution for high-reliability applications.

The unique properties of GaN transistors, specifically High Electron Mobility Transistors (HEMTs), make them particularly suited for challenging environments. These devices can maintain performance where other semiconductors would fail, offering significant advantages in space electronics, satellite communications, and high-power defense systems. Researchers have found that GaN-based devices show an order of magnitude more radiation hardness compared to traditional semiconductor technologies.

The nuclear interactions between the particle and the atom cause displacement damage. When an incoming particle collides with a lattice atom, the atom is displaced from its lattice position, resulting in lattice defects. However, it reduces the carrier lifetime and causes the device's reliability to be disrupted. There are various sources of radiation from which displacement of atoms happen and disrupt the efficient functionality of the device. This section discusses the displacement effects due to lattice disorder and scattering happening in HEMT devices as a result of being subject to several sources of irradiation. Depicts the formation of displacement damage effect caused in a GaN device due to nuclear interactions.

As shown, the proton irradiated particle knocks over an atom to form an interstitial atom, thus creating a vacancy. The primary radiated particle gets scattered out of the lattice after reflecting. The interstitial and vacancy atom form a Frenkel pair. For instance, throughout the mission life of a spacecraft, the total power output of a solar array diminishes due to displacement effects. The self-heating effects caused as a result of the change in thermal boundary resistance at a proton fluence of $2 \times 10^{15} \text{ cm}^{-2}$, showed exacerbated performances with less power density.

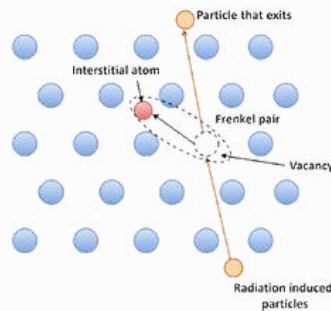


Fig 3: Schematic illustration of displacement damage caused in a GaN crystal structure upon proton irradiation. Lattice defects originate from atomic displacements, resulting in localized trap states

Single event ion strikes on AlGaIn/GaN devices can cause a new type of damage. This damage creates leakage paths between the source and drain regions and also leads to more charge buildup from atom displacement. Lower levels of ion strikes (around $1\text{E}7 \text{ cm}^{-2}$) usually do not cause permanent damage. However, at higher levels (above $1\text{E}10 \text{ cm}^{-2}$), heavy ions can move atoms in the silicon structure. This can permanently degrade device performance, resulting in shifts in threshold voltage and higher leakage currents. Even at low levels, permanent damage can occur through processes like carrier trapping. The type of heavy ion, such as protons or argon ions, also matters. Heavier ions with more energy can cause greater damage than lighter ions at the same levels. At low levels, a single high-energy heavy ion can deliver enough energy to trigger a condition called Single Event Burnout (SEB) in a High Electron Mobility Transistor (HEMT), especially if it hits a weak spot in the device. While it is unlikely for SEB to happen from just one low-level event, repeated exposures, even at low levels, can lead to gradual damage and eventual device failure. Abbate et al. (2015) studied the gate leakage from SEB. They pointed out that heavy ion doses cause charge amplification, but this also leads to increased leakage. Their research shows how the risk of SEB can make these devices less suitable for use in space.

In the process of identifying constraints associated with Single Event Effects (SEE) and pinpointing the sensitive volume within the device, a pronounced variation in the residual electric field emerged, influenced by trap density during irradiation simulations. The commercial normally-off GaN switch demonstrated its capability to function under diverse conditions of heavy-ion irradiation, which included various factors such as the position of ion tracks, their range, and the bias applied between the source and drain terminals. A detailed illustration in captures the moment heavy ions collide with the GaN device, revealing that the ion beam makes its way through the depleted region, specifically near 2DEG. This interaction significantly extends the depletion region in alignment with the magnitude of the ion beam, leading to the generation of electron-hole pairs and resulting in an altered potential contour distribution throughout the device. Such disruptions not only impact the energy band structure but also compromise the overall performance of the device. Upon striking the device, the heavy ion leads to a noticeable decline in the electric field strength after it penetrates the field-plate layer and reaches the buffer region. This initial pulse of ion-induced radiation can foster charge accumulation at the channel, presenting limitations on the choice of channel materials suitable for space-based applications due to the variability of these effects. Furthermore, the potential drop that occurs across the various access regions contributes to additional charge accumulation near the gate terminal. A considerable volume of charge is ultimately deposited beneath the depleted region of the GaN transistor. Here, the electric field reaches a magnitude sufficient to generate high current drives, which can instigate electrical distortions and potentially cause irreversible damage to the device. These intricate interactions and their consequences are essential considerations for the Device Under Test (DUT), highlighting the significant challenges posed by heavy-ion irradiation.

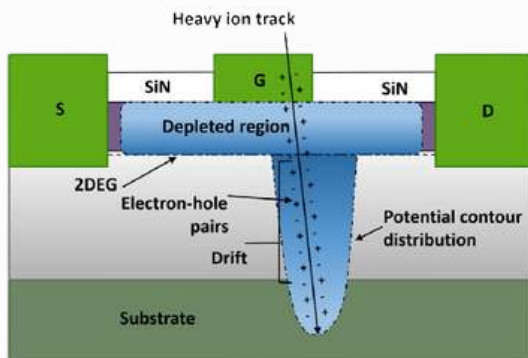


Fig4: Schematic representation of the heavy-ion track impact on GaN transistor

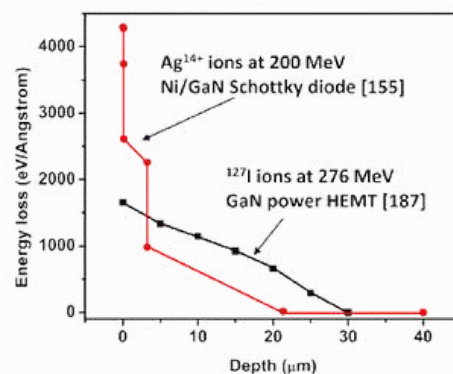


Fig 5:SRIM simulated depiction of energy loss in 127I [187] and Ag14+ [156] ions

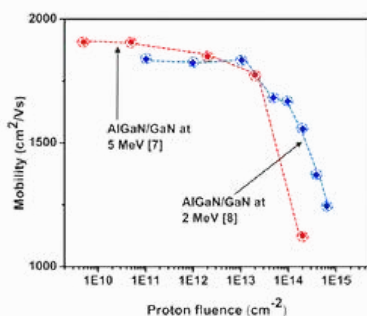


Fig6: Field-effect mobility deterioration of several devices over the years due to alloy disorders with respect to proton radiation fluences.

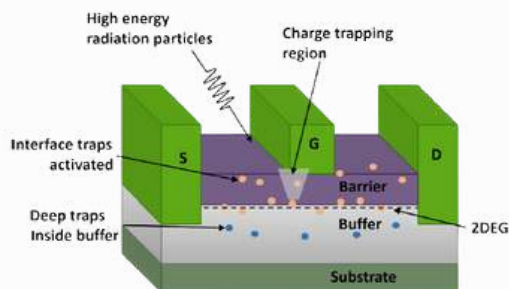


Fig 7: Traps formed at the interface and 2DEG regions due to irradiation.

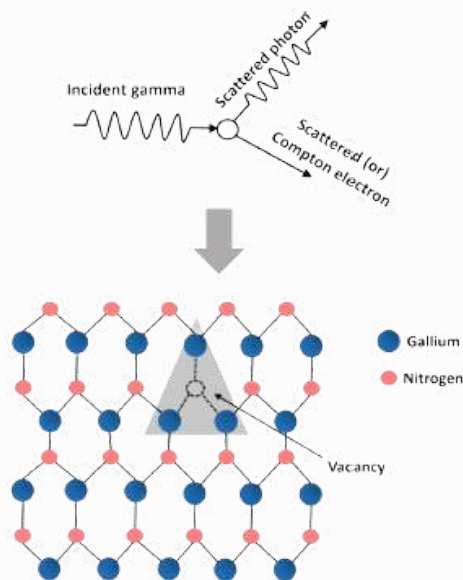


Fig7: Nitrogen vacancy formation in GaN crystals upon gamma radiation through compton scattering. Ga atoms need to reoccupy substitutional positions after radiation and subsequent annealing to repair the lattice.

The resilience of GaN isn't just about surviving radiation—it's about maintaining performance. Even under intense radiation exposure, these transistors can preserve critical electrical characteristics like carrier mobility, threshold voltage, and overall device functionality. This makes them invaluable in applications where failure is not an option, such as spacecraft electronics, radar systems, and advanced communication networks.

As technology pushes the boundaries of what's possible in extreme environments, GaN transistors stand at the forefront of innovation, promising more reliable, durable, and high-performance electronic solutions for the most demanding applications.



Dr. J. Charles Pravin
Professor, ECE

PERFORMANCE ESTIMATION OF MULTILAYER-STACK-CHANNEL IGZO BASED THIN-FILM TRANSISTOR IN DOUBLE-GATE MODE

Several additional amorphous Indium-Gallium-Zinc Oxide (IGZO) have been employed as a promising material in Thin Film Transistors (TFT) applications due to its good transparency, resilience, and cost-effectiveness. However, due to low attainable mobility and low turn-on currents, the single layer channel IGZO TFT is insufficient. To improve the switching characteristics of IGZO TFTs, this research suggests a multi-stack channel configuration. In this study, the source of performance enhancement in Double-Gate (DG) -IGZO TFTs when compared to the Single-Gate (SG) multilayer stack channel structure is explained. Low mobility and Subthreshold Swing (SS) degradations occur in SG multilayer stack channel TFTs. The field-effect mobility (FE) of SG TFTs ($14.70 \text{ cm}^2/\text{V}\cdot\text{s}$) increases to $24.92 \text{ cm}^2/\text{V}\cdot\text{s}$ when the gate voltage ($V_G = 10 \text{ V}$) is increased. From the channel's top to the gate-insulator interface ($1.3 \times 10^{18} \text{ cm}^{-3}$), the concentration of electrons in SG TFT declines gradually ($1.86 \times 10^{18} \text{ cm}^{-3}$). The current density in the DG structure of the multi-stack channel drops from 7.5 A per cm^2 to 2.5 A per cm^2 , leading in accumulation in the -IGZO bulk and a large drain current. When the results from the proposed structure were compared to current results, they were found to be extremely encouraging and promising for future displays.

Introduction: Zinc Oxide (ZnO) is a promising material for Transparent Conducting Oxides (TCOs) due to its wide-ranging control of electrical and optical properties, high transparency, and moderate mobility, making it ideal for optoelectronics applications. Over the past two decades, ZnO-based Thin-Film Transistors (TFTs) have been developed for next-generation flexible and 3D displays, owing to their high mobility, scalability, and low-temperature processing. TFTs, similar to Metal-Oxide-Semiconductor (MOS) transistors, have an insulating substrate and undoped channel, but their operation mechanism differs. Among various types, α -IGZO (amorphous-Indium Gallium Zinc Oxide) TFTs stand out for their transparency, electrical characteristics, and durability, finding applications in OLEDs, UHD, LCDs, and sensors. Despite their advantages, the single-active layer channel in α -IGZO TFTs suffers from low film density and mobility limitations. To address this,

multi-stack channel designs using materials like ITO, IZO, and AIZO have been proposed, with some studies focusing on multi-layer stacks and double-gate control structures to enhance performance.

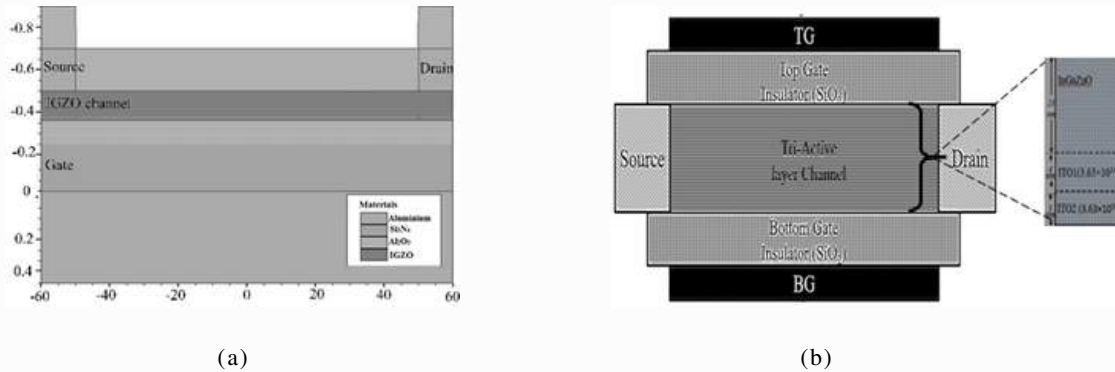


Figure 1: Schematic of (a) SG TFT structure (b) DG-multi-stack TFT structure

Zinc Oxide (ZnO) and α -IGZO materials are promising for transparent electronics, offering high mobility, scalability, and low-temperature processing, particularly in Thin-Film Transistor (TFT) applications for displays and sensors. The inclusion of Gallium (Ga) in α -IGZO suppresses oxygen vacancy formation, improving the electrical performance of the material by reducing off-current and sub-threshold swings, though at the cost of mobility. This research explores multi-layer active channel structures to enhance device performance, proposing a dual-active layer with varying Ga content to achieve a balance between mobility and electrical characteristics. The study compares single-gate (SG) and double-gate (DG) IGZO TFTs, showing that the DG structure improves field-effect mobility and achieves a higher I_{on}/I_{off} ratio (~ 107) while maintaining a low sub-threshold swing (~ 158 mV/decade). The results indicate that multi-layer stacking of materials and optimizing channel structure lead to significant performance improvements, making the proposed multi-stack designs promising for next-generation optoelectronics

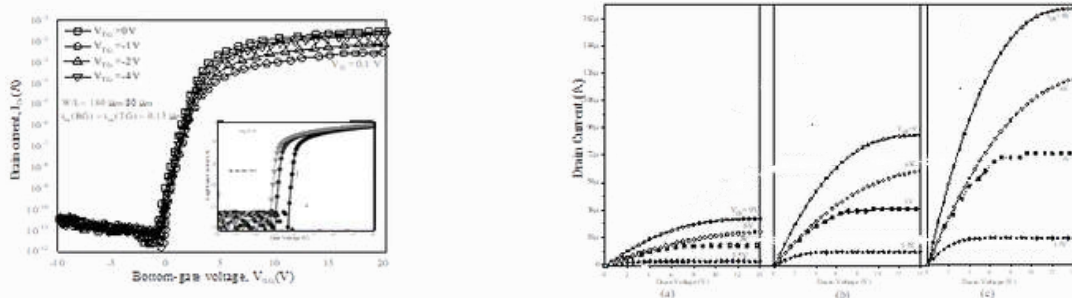


Figure 2 (a): Transfer-characteristics (I_D - V_G) of SG and DG-TFTs., (b) Comparison of drain characteristics TAL IGZO TFT (a) Single-Gate Bottom, (a) Single-Gate Top, and (b) Double-Gate TFT.

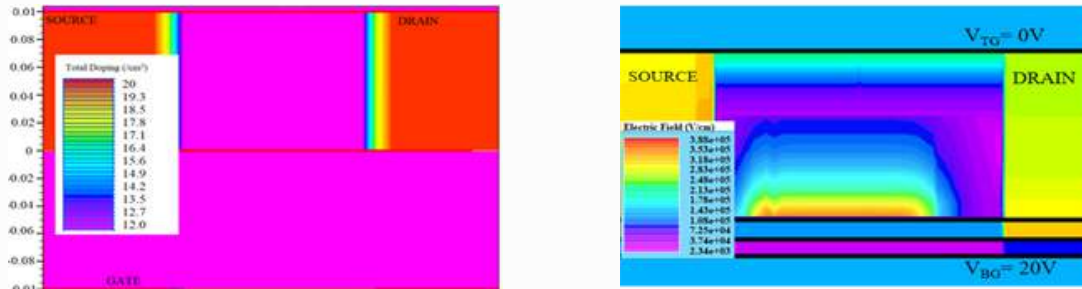


Figure 3 (a). SG-IGZO TFT Doping Distribution, (b) Fig. 8. Electric Field distribution in DG-multi-stack IGZO TFT

The cause of performance enhancement in the double-gate -IGZO compared to the single-gate architecture is described in this paper. Mobility and SS degradations have caused anguish in the SG TFTs. The field-effect mobility (FE) of single-gate TFTs is 14.70 cm²/V-s, whereas the FE of double-gate TFTs is 24.92 cm²/V-s. From the channel's top to the gate-insulator contact (1.3 10¹⁸ cm⁻³), the density of electrons in SG TFT decreases steadily (1.861108 cm⁻³). The current density of the DG TFT reduces from 7.5 A/cm² to 2.5 A/cm² for similar scenarios, creating buildup in the a-IGZO bulk and resulting in a greater drain current. The DG-IGZO TFT can replace the high mobility of SG TFTs in array circuits such as Flexible RFIDs, ultra-HD displays, and wearable electronics. However, improving subthreshold swing (SS) persists in DG and SG, both IGZO TFT architectures, and would necessitate more study and structural changes.



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Student Articles

NEURALINK

Introduction:

Neuralink is a neurotechnology company founded by “Elon Musk” and others in 2016. Its primary goal is to develop implantable brain-computer interfaces (BCIs) with the aim of enhancing human capabilities and treating neurological conditions.

What Is Neuralink?

Neuralink is a device, specifically, a Brain Machine Interface, that will be surgically implanted into the brain and with it which can be able to communicate with machines and even control them.

Why Neuralink?

- Helps people with severe spinal cord injury by giving them the ability to control computers and mobile.
- Restoring motor and sensory function and the treatment of neurological disorders.
- A Preserve and enhance your own brain.
- A Security will be built into every layer of the product, using strong cryptography, defensive engineering, and extensive security auditing.



Overview of Neuralink:

- The Link processes, stimulates, and transmits neural signals.
- A Surgically transplanted inside head.
- A Micron-scale threads are inserted into areas of the brain that control movement.
- A Compact inductive charger wirelessly connects.
- It has potential to treat brain disorders and other medical problems.
- A Give us the power to interact with and control machines using our minds

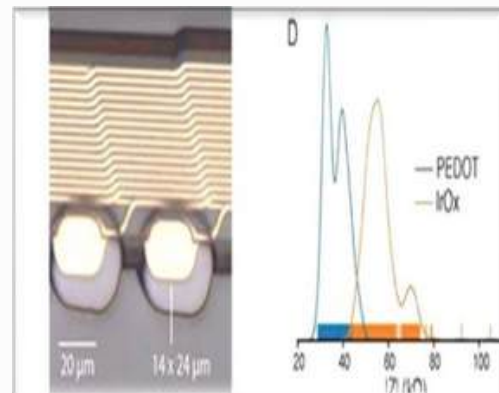
Electronics in Neuralink:

- Application specific integrated circuit (ASIC) is used.
- A number of ASICs are integrated into a standard printed circuit board (PCB).
- Each system consists of a field programmable gate array (FPGA)
- Real-time temperature
- Accelerometer
- Magnetometer sensors
- A single USB-C connector
- The systems are packaged in titanium cases which are coated with parylene-c.



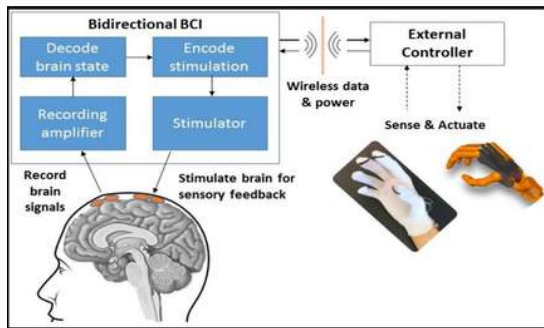
Neural Thread:

- Each small and flexible thread contains many electrodes for detecting neural signals.
- A Thread thickness is nominally 4 to 6 μm .
- A Typical thread length is approximately 20 mm.
- A The main substrate and dielectric used in these probes is polyimide.



Working:

Neuralink, founded by Elon Musk, aims to develop brain-computer interface (BCI) technology to merge the human brain with artificial intelligence (AI) systems. Here are some of the working principles of Neuralink:



Implantable Devices: Neuralink creates small, implantable devices that interface with the brain's neurons. These devices, often referred to as "neural lace," are designed to be minimally invasive and to interface with the brain's neurons for both input and output.

Electrodes: The core component of Neuralink's technology is an array of electrodes that are embedded into the brain. These electrodes are capable of both recording neural activity and stimulating neurons.

Neural Recording: The electrodes can detect electrical signals from neurons, allowing for the recording of brain activity. This data can be analyzed to understand how the brain processes information and generates thoughts.

Neural Stimulation: Neuralink's electrodes can also stimulate neurons, potentially enabling the brain to receive input from external sources or to bypass damaged areas of the brain.

Data Processing and Interpretation: The signals recorded by Neuralink's electrodes are processed by advanced algorithms to interpret the neural activity. This processed data can be used to control external devices, such as computers or prosthetic limbs, through thought alone.

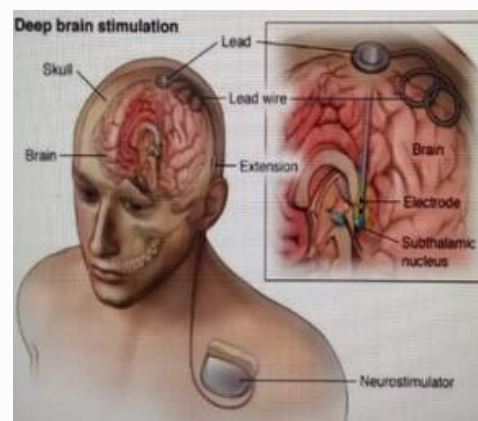
Wireless Communication: Neuralink devices are designed to wirelessly communicate with external devices, such as smartphones or computers. This enables seamless integration of the brain with digital technology.

Machine Learning: Neuralink's system employs machine learning algorithms to improve its ability to interpret and respond to neural signals over time. This adaptive approach allows for more accurate and efficient interaction between the brain and external devices.

Biocompatibility and Safety: Ensuring the safety and biocompatibility of Neuralink's devices is crucial. The company utilizes advanced materials and surgical techniques to minimize the risk of rejection or damage to the brain tissue.

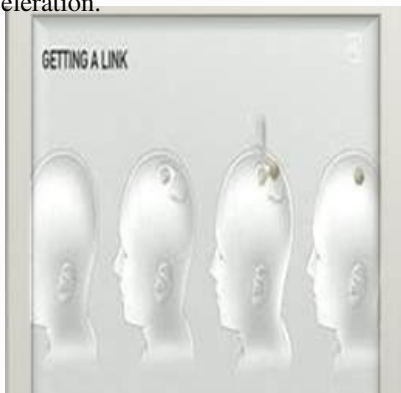
How Neuralink is different from other BMI devices:

- There are currently only a few approved devices for recording or stimulating from the human brain.
- These are designed to modulate the activity of whole brain areas, not to transfer information to and from the brain.
- The Link has more electrodes and with flexible threads.
- The Link will also perform real-time spike detection on every channel.



How it is placed in the body?

- A robotic system will be used.
- The neurosurgical robot capable of inserting six threads per minute.
- A Have an auto-insertion mode feature.
- Each thread can be individually inserted into the brain with
- Micron precision.
- A The needle is driven by a linear motor allowing variable insertion speeds and rapid retraction acceleration.



Challenges in making scalable BMI

- Scale up the number of electrodes while also building a safe and effective clinical system that users can take home and operate by themselves.
- The threads have to resist corrosion from fluid in the tissue, and the electrodes must have sufficient surface area to allow stimulation.
- Scaling these devices requires on-chip, real-time identification and characterization of neural spikes.
- Design adaptive algorithms that maintain reliable and robust performance.

Current status of Neuralink:

As of my last update in January 2022, Neuralink has made significant progress but has not yet achieved widespread commercial availability of its brain-computer interface (BCI) technology. Here are some key updates on the current status of Neuralink:

- **Surgical Implants in Animals:** Neuralink has conducted experiments with its BCI technology in animals, including rats and pigs. These experiments have demonstrated the feasibility of implanting Neuralink devices into the brain and recording neural activity.

- **Public Demonstrations:** Neuralink has hosted several public demonstrations showcasing its technology and providing updates on its progress. These demonstrations have included live demonstrations of Neuralink implants in animals, as well as presentations on the company's goals and research.

Applications of Neuralink:

Medical Treatments: Neuralink has the potential to transform the treatment of neurological conditions like Parkinson's disease, epilepsy, and depression through its precise brain stimulation and recording capabilities.

Assistive Technology: It could offer assistive technology for individuals with disabilities, allowing for direct brain control of prosthetic limbs, wheelchairs, or other assistive devices.

Communication: Neuralink might enable new forms of communication, including direct brain-to-brain communication, facilitating communication for individuals with speech or motor impairments, and potentially leading to advanced telepathic communication methods.

Cognitive Enhancement: Neuralink's technology could augment memory, learning, and cognitive abilities through direct brain-computer interfaces, potentially leading to cognitive enhancement for both medical and non-medical applications.

Entertainment and Gaming: Neuralink's technology might enable immersive virtual reality experiences and gaming interfaces controlled directly by the brain, offering a new level of immersion and interaction.

Research and Understanding: It could advance our understanding of the brain and neurological processes, leading to breakthroughs in neuroscience, cognitive science, and the treatment of neurological disorders.

Conclusion:

Neuralink has developed advanced solution to existing medical problems. Their focus on development, expansion, and regulatory compliance highlights their commitment. With ongoing efforts, Neuralink aims to revolutionize neuroscience and potentially assist individuals with neurological conditions. However, commercial availability and broader impact remain pending as the company navigates the complexities of bringing its vision to reality.

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LIFE ON MARS



DISCOVERING MARS LIFE BEYOND EARTH..?

Welcome to the inaugural issue of Red Horizon, your premier source for all things related to life on Mars! As we stand on the precipice of a new era in space exploration, this magazine aims to captivate your imagination, provide insights into the scientific marvels of the Red Planet, and envision the lifestyle of future Martian inhabitants. Mars, the fourth rock from the sun, has long fascinated humanity. In recent years, advancements in space exploration and technology have brought us closer to unlocking the mysteries of this enigmatic world. Red Horizon is your passport to the future, a journey that explores the possibilities and challenges of life on Mars

As of my last update in January 2022, there has been no confirmed discovery of life beyond Earth, including on Mars. However, there have been significant efforts by space agencies and researchers to explore Mars for signs of past or present life.

NASA's Perseverance rover, which landed on Mars in February 2021, is equipped with advanced scientific instruments designed to search for signs of ancient microbial life. Perseverance is exploring Jezero Crater, which scientists believe was once flooded with water and may have preserved evidence of past microbial life.

The focus on Mars for potential human colonization arises from a combination of factors that make it more suitable compared to other planets in our solar system.

Presence of Resources: Mars is believed to have various resources that could be valuable for sustaining human colonies, including minerals, metals, and potentially usable subsurface water. The utilization of local resources is crucial for long-term sustainability.



While Mars has unique advantages, it's essential to note that exploring and understanding other celestial bodies in our solar system and beyond continues to be a priority in space exploration. Each planet or moon presents its own set of challenges and opportunities, and future missions may expand our understanding of potential destinations for human exploration and colonization

Similar Day-Night Cycle: Mars has a day-night cycle similar to Earth, with a day lasting approximately 24.6 hours. This similarity can ease the adaptation for humans in terms of circadian rhythms, daily routines, and the utilization of solar energy.

LATEST DISCOVERIES ON MARS



1. Mars Exploration: Ongoing missions, such as NASA's Perseverance rover, continue to explore Mars. Perseverance landed on Mars in February 2021 and is conducting experiments, including the search for signs of past life and the production of oxygen from the Martian atmosphere. The Zhurong rover, part of China's Tianwen-1 mission, has been active on Mars, conducting scientific studies and exploring the Utopia Planitia region.

2. Methane Detection: Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur a tempus risus. Nulla tristique libero quis neque sagittis, ac lacinia turpis rutrum. Praesent vulputate vulputate ligula, a condimentum enim sodales vitae.

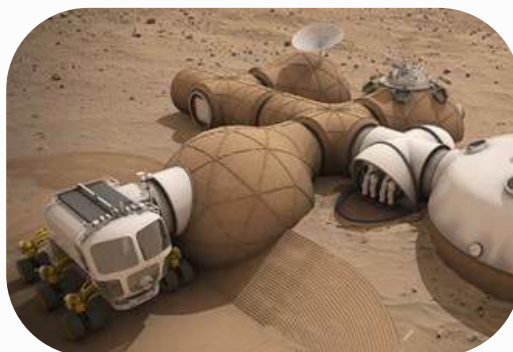
3. Ancient Lakes and Rivers: Earlier Mars rovers, such as Curiosity, identified geological features suggesting the presence of ancient lakes and rivers. These findings support the hypothesis that liquid water was once more abundant on the Martian surface.

4. Tianwen-1 Mission (China): China's Tianwen-1 mission, which includes the Zhurong rover, successfully landed on Mars. Zhurong is conducting scientific studies and exploring the Utopia Planitia region.

unleash your imagination as we showcase avant-garde architectural designs tailored for the unique challenges of Mars. From sleek and compact living spaces to awe-inspiring colony concepts, we explore the creative minds shaping the skyline of the first Martian cities. Discover the harmony between form and function as architects redefine what it means to build on an alien world

FUTURE ARCHITECTURES

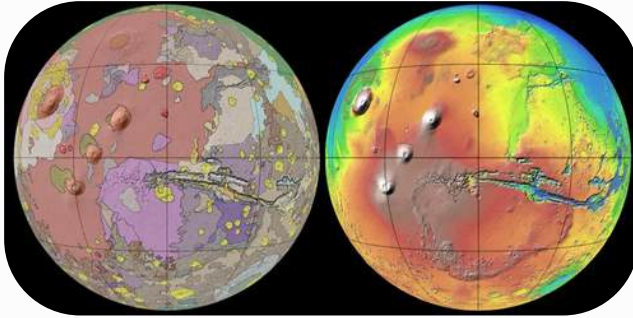
1. "Martian living" refers to the lifestyle and conditions associated with living on Mars. This concept encompasses various aspects, including the design of habitats, daily routines, social interactions, and the adoption of technologies and practices tailored to the unique challenges and environment of the Red Planet. Essentially, it explores how humans might adapt and thrive in a hypothetical community on Mars, considering factors such as reduced gravity, limited resources, and the need for sustainable and innovative solutions.



2. On Mars, sustainability isn't just a choice; it's a necessity. Join us in unraveling the innovative practices that could sustain life on the Red Planet. From cutting-edge recycling systems to renewable energy sources, we investigate the technological marvels that will form the backbone of a self-sufficient Mars community. Explore the delicate balance between human needs and environmental preservation, paving the way for a sustainable Martian future.



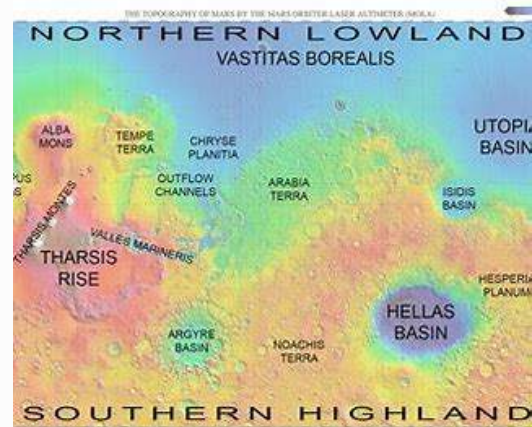
MARS GEOGRAPHY



As it is known, Mars is the fourth and the last of the inner and rocky planets of our Solar system. Beyond the “Red Planet”, it extends the belt of asteroids just before the giant Giove and the other gaseous external planets of the system. It has a mass: 6.417×10^{23} Kg, a density of $3,940 \text{ g/cm}^3$, equatorial diameter of 6,792 Km and a mean temperature of -63°C , a distance from Sol of $228,000,000 \text{ km}$. Mars orbits the Sun at an average distance of 230 million km and its revolution period is about $687 \times \text{days}$; while his solar day is a little longer than ours: $24 \times \text{hours}$, $37 \times \text{minutes}$ and $23 \times \text{seconds}$. The Martian axial inclination is 25.19° , which is similar to that of Earth. Due to the discrete eccentricity of its orbit of 0.093, its distance from Earth to opposition can range between about 100 and about 56 million kilometers; only Mercury has a higher eccentricity in the Solar System. However, Mars used to follow a much more circular orbit: about 1.35 million years ago its eccentricity was equivalent to 0.002, which is much lower than the current Earth's. Mars has an eccentricity cycle of 96,000 Earth years compared to Earth's 100,000. Over the past 35,000 years, Martian orbit has become increasingly eccentric due to the gravitational influences of other planets, and the closest point between Earth and Mars will continue to decline over the future time.

Mars, often referred to as the “Red Planet,” holds a special place in our solar system as the last of the inner rocky planets before transitioning to the vast expanse of the outer gas giants. With its intriguing characteristics, such as a mass of 6.417×10^{23} kg, a density of $3,940 \text{ g/cm}^3$, and an equatorial diameter of 6,792 km, Mars never fails to captivate the imagination of space enthusiasts. Mars, often referred to as the “Red Planet,” holds a special place in our solar system as the last of the inner rocky planets before transitioning to the vast expanse of the outer gas giants.

With its intriguing characteristics, such as a mass of 6.417×10^{23} kg, a density of $3,940 \text{ g/cm}^3$, and an equatorial diameter of 6,792 km, Mars never fails to captivate the imagination of space enthusiasts.



The environmental and geological history of early Mars is mostly written in its rocks, in their composition and morphological/structural signatures. The first few billion years of Mars' geologic history records surface environments considerably different than the surface today, prompting a succession of coordinated surface and orbiter missions over the past two decades aimed ultimately at determining if Mars ever had an early biosphere. Water is the sine qua non for all life as we understand it and, therefore, past missions have sought environments where water was abundant and possibly long-lived [1, 2]. Mars was formed 4.6 billion years ago, with a history similar to the other three Terrestrial-type planets, i.e. as a result of the condensation of the solar nebula, most probably silicates.

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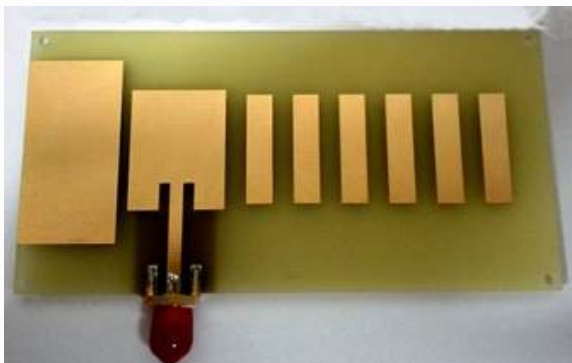
MICROSTRIP ANTENNAS

Introduction:

Microstrip Antenna was initially proposed by G.A. Deschamps in year 1953 but came into existence in 1970's when Robert E. Munson and some of his fellow researchers developed it using a low loss substrate. Also known as Patch Antenna. These Antennas are mainly used at microwave frequency (above 1 GHz).

What is a Microstrip Antenna?

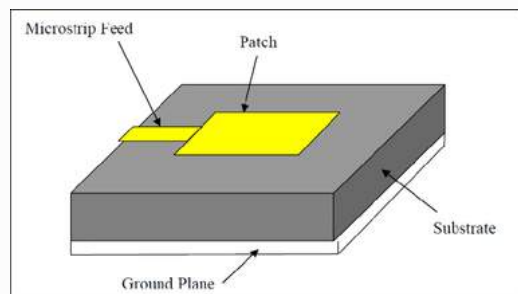
An antenna that is shaped by simply etching out a piece of conductive material above a dielectric surface is called a microstrip antenna or a patch antenna. On the ground plane of this microstrip antenna, the dielectric material is mounted, where this plane supports the entire structure. In addition, the excitation to this antenna can be provided with feed lines that are connected to the patch. Generally, these antennas are considered low-profile antennas that are used in microwave frequency applications that have above 100 MHz frequency.



Antenna's micro-strip/patch can be selected to be rectangular, square, elliptical & circular for ease of analysis and fabrication. Some microstrip antennas do not utilize a dielectric substrate but they are made with a metal patch that is mounted on a ground plane with dielectric spacers; thus the resulting formation is less strong but its bandwidth is wider.

Construction of Microstrip Antenna:

Microstrip antenna design can be done with the help of an extremely thin metallic strip by arranging it on a ground plane in between a dielectric material. Here, the dielectric material is a substrate used for separating the strip from the ground plane. Once this antenna is excited, then the generated waves in the di-electric undergo reflections & the energy emitted from the metal patch edges is very low. These antenna shapes are identified by the metallic patch shape arranged on the dielectric material.



The strip/patch & the feed lines are photo-etched on the surface of the substrate. There are different microstrip antenna shapes like square, dipole, rectangular, circular, elliptical, and dipole. We know that patches can be formed in various shapes but, because of the easy in fabrication, circular, square & rectangular shaped patches are normally used. Microstrip antennas can also be formed with a group of various patches above a dielectric substrate. Either single or numerous feed lines are utilized to give excitation to the microstrip antenna. So the presence of microstrip element arrays provides better directivity, high gain, and increased range of transmission with low interference.

Working of microstrip antenna:

A microstrip antenna works as; whenever current throughout a feed line arrives at the microstrip antenna's strip, then electromagnetic waves are produced. So, these waves from the patch will start radiating from the width side. However, when the strip thickness is very small, the waves produced in the substrate will get reflected through the strip edge. The constant strip structure along the length does not allow radiation emission. The microstrip antenna's low radiating capability allows for covering only wave transmissions with small distances like stores, indoor locations, or local offices. So, this inefficient wave transmission is not acceptable in a centralized locality in a very large area. Usually, hemispherical coverage is given by a patch antenna at a 30° 180° angle at a distance from the mount.

Microstrip Antenna Specifications:

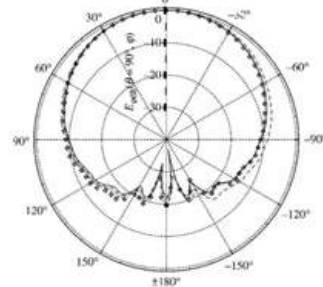
The microstrip antenna specifications include the following

- Its resonant frequency is 1.176 GHz.
- The frequency range of the microstrip antenna is from 2.26 GHz to 2.38 GHz.
- The substrate's Dielectric constant is 5.9.
- The dielectric substrate's height is 635 μm .
- The feeding method is a microstrip line feed.
- The loss tangent is 0.0012.
- The conductor is silver.
- The thickness of the conductor is 25 μm .
- Its Bandwidth is $f_0 \pm 10$ GHz.
- Its gain is above 5 dB.
- Its axial ratio is below 4 dB.
- Its return loss is better than 15 dB.

Microstrip Antenna Radiation Pattern:

The graphical representation of the antenna's radiation properties is known as radiation pattern which explains how the antenna emits energy into space. The variation in the power as an arrival angle's function is monitored in the far field of the antenna.

The microstrip antenna radiation pattern is broad and it has less radiation power & narrow frequency BW. The microstrip antenna's radiation pattern is shown below which lesser directivity has. By using these antennas, an array can be formed to have a superior directivity.

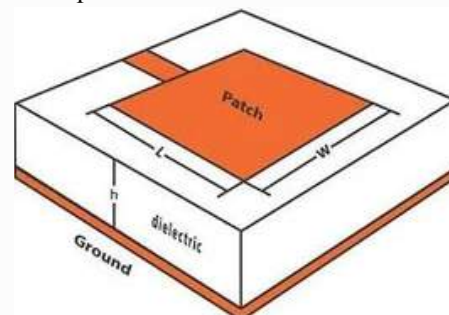


Microstrip Antenna Types:

There are different types of microstrip antennas available. They are;

Microstrip Patch Antenna

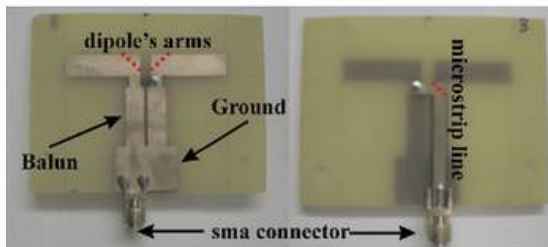
These types of antennas are low-profile antennas where a metal patch is arranged at ground level through a dielectric material in-between comprising a strip (or) Patch Antenna. These antennas are extremely low size antennas with low radiation. This antenna includes a radiating patch on one face of a dielectric substrate and on the other side, it has a ground plane.



Generally, the patch is made with conducting material like gold or copper. These types of antennas can be formed with a microstrip method by simply fabricating on a PCB. These antennas are used in microwave frequency applications which have greater frequency than 100 MHz.

Microstrip Dipole Antenna

The microstrip dipole antenna is a thin microstrip conductor and is placed on the actual part of the substrate & it is covered totally with metal on one face known as the ground plane. These antennas are used in digital communication devices like computers & the nodes for WLAN. The width of this type of antenna is small so it can be utilized at the entrée point of the WLAN system.



Microstrip Travelling Wave Antenna:

Microstrip traveling wave antennas are mainly designed with a long Microstrip line by sufficient width for supporting the TE connectivity. These types of microchip antennas are designed in such a way that the major beam lies within any route from broadside to end fire.



Characteristics of microstrip antenna

The microstrip antenna patch should be an extremely thin conductive region.

As compared to a patch, the ground plane should have fairly extremely large dimensions.

Photo-etching on the substrate is done to construct the radiating element & feed lines.

A thick dielectric substrate by the dielectric constant in the 2.2 to 12 range offers excellent performance of an antenna.

Advantages and disadvantages of microstrip antennas:

Advantages of microstrip antenna

- Microstrip antennas are very small.
- These antenna's weight is less.
- The fabrication procedure provided by this antenna is simple.
- Its installation is very easy because of its small size & volume.
- These antenna arrays can be constructed easily.

Disadvantages of microstrip antenna

- This antenna provides less gain.
- The efficiency of this type of antenna is low due to conductor & dielectric losses.
- This antenna has a high range of cross-polarization radiation.
- It has less impedance bandwidth.

Applications of microstrip antennas:

- Microstrip antennas are applicable in different fields, in missiles, satellites, space crafts, aircraft, wireless communication systems, mobile phones, remote sensing & radars.
- These antennas are used in wireless communications to show compatibility with handheld devices such as mobile phones & pagers.
- These are used on missiles as communication antennas.
- These antennas have a small size, soused in microwave and satellite communication applications.
- GPS is one of the main benefits of microstrip antennas because it provides ease within vehicles & marines tracking.

Conclusion

Microstrip antennas represent a significant advancement in antenna technology, offering a balance between performance, size, and ease of integration. As wireless communication systems evolve and demand for compact, efficient antennas grows, microstrip antennas are poised to remain a crucial component in the ever-expanding landscape of modern communication technologies.

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AI-POWERED IMAGE PROCESSING: A TECHNICAL ODYSSEY

Abstract:

In today's technological landscape, Artificial Intelligence (AI) stands as a pioneering force, reshaping industries and revolutionizing human-machine interaction. Within this paradigm, AI's fusion with image processing emerges as a cornerstone, unlocking novel possibilities across diverse domains. This article delves into the intricate realm of AI-enhanced image processing, elucidating core principles, cutting-edge advancements, practical applications, and ethical considerations.

Introduction:

In recent years, the fusion of artificial intelligence (AI) and image processing has catalyzed transformative advancements across various sectors. This article delves into the technical intricacies of AI-driven image processing, elucidating its underlying principles, cutting-edge techniques, and burgeoning applications. From convolutional neural networks (CNNs) revolutionizing image recognition to generative adversarial networks (GANs) pushing the boundaries of image generation, this article navigates the landscape of AI in image processing with a focus on its profound implications and ethical considerations.

The convergence of AI and image processing represents a watershed moment in technological innovation, heralding a new era where machines interpret and manipulate visual data akin to human cognition. This article provides a comprehensive exploration of the technical foundations, advancements, and ethical dimensions of AI-powered image processing.



Understanding Image Processing with AI

At its core, image processing with AI entails the utilization of algorithms to analyze, interpret, and manipulate visual data. Leveraging deep learning architectures such as CNNs, AI systems can extract intricate features from images, enabling tasks such as object detection, pattern recognition, and image classification with unprecedented accuracy and efficiency.

AI and Image Processing Basics

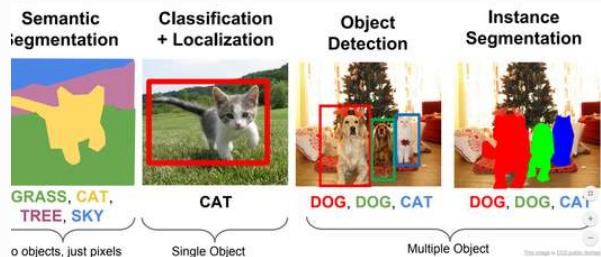
Image processing, bolstered by AI, encompasses a multifaceted approach, spanning acquisition, preprocessing, feature extraction, analysis, and post-processing stages. The amalgamation of traditional techniques with AI algorithms heralds a new era of visual data interpretation and manipulation.

Revolutionizing with AI

AI's ascendancy, particularly in the realm of deep learning, has catalyzed a paradigm shift in image analysis. Through the iterative process of learning from vast datasets, AI systems exhibit unparalleled proficiency in discerning intricate patterns, facilitating real-time processing and adaptive task execution.

Historical Evolution

Tracing the evolutionary trajectory of AI in image processing unveils a narrative punctuated by seminal milestones. From rudimentary rule-based systems to the epoch-making advent of deep learning architectures, each phase marks a leap forward in computational capability and cognitive prowess.



Understanding Image Processing with AI

At its core, image processing with AI entails the utilization of algorithms to analyze, interpret, and manipulate visual data. Leveraging deep learning architectures such as CNNs, AI systems can extract intricate features from images, enabling tasks such as object detection, pattern recognition, and image classification with unprecedented accuracy and efficiency.

Applications Across Industries

Healthcare: AI aids in medical imaging, facilitating diagnosis and treatment planning, and unveiling insights that were erstwhile inaccessible.

Automotive: Autonomous vehicles rely on AI-driven image processing for safe navigation, while Advanced Driver-Assistance Systems (ADAS) bolster driver awareness.

Entertainment: The entertainment landscape harnesses AI in content creation, from enriching visual effects to sculpting immersive virtual reality escapades.

Advancements in AI Techniques

Convolutional Neural Networks (CNNs) have emerged as a cornerstone of AI-powered image processing, revolutionizing tasks such as image recognition and segmentation. Through the intricate interplay of convolutional layers, pooling operations, and nonlinear activation functions, CNNs exhibit remarkable prowess in discerning intricate visual patterns.

Generative Adversarial Networks (GANs) represent another breakthrough in AI-driven image processing, facilitating the generation of photorealistic images from textual descriptions.

By pitting a generator network against a discriminator network in a competitive framework, GANs engender a dynamic interplay that converges towards the production of increasingly realistic images.

Transfer learning has emerged as a pivotal technique in AI-powered image processing, enabling models to leverage knowledge gained from one domain to expedite learning in another. By fine-tuning pre-trained models on domain-specific data, transfer learning mitigates the need for vast labeled datasets, thereby streamlining the model development pipeline.

Conclusion:

The fusion of AI and image processing heralds a new frontier of technological innovation, with far-reaching implications across industries and societal domains. By elucidating the technical underpinnings, advancements, and ethical considerations of AI-powered image processing, this article underscores the imperative of responsible AI development and deployment. As we navigate this transformative landscape, a proactive approach to addressing ethical challenges and fostering equitable access to AI-driven solutions is paramount to realizing the full potential of this paradigm-shifting technology.



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GENETICALLY ENGINEERED BABIES

Genetically Engineered Babies: A Pandora's Box of Potential?

The human desire to shape our offspring is as old as civilization itself. From carefully planned marriages to prenatal vitamins, we've constantly sought ways to influence the next generation. Now, with the advent of CRISPR-Cas9 and other gene-editing tools, we stand at the precipice of a new era – one where we can directly alter the human genome before birth. This article delves into the world of genetically engineered babies, exploring the science, ethics, and societal implications of this nascent technology.

Why Consider Engineering Our Babies?

The primary motivation for this technology lies in eradicating disease. Thousands of debilitating genetic conditions, from cystic fibrosis to Tay-Sachs disease, are passed down through generations. Gene editing offers the possibility of eliminating these faulty genes, giving children a healthier life free from the burden of inherited illness. Additionally, it could potentially enhance resistance to infectious diseases or even modify certain traits associated with higher susceptibility to chronic illnesses like heart disease or diabetes.



A Look Back: A Timeline of Gene Editing

The journey towards engineered babies began with the discovery of the DNA double helix in 1953. The 1970s saw the birth of recombinant DNA technology, enabling the manipulation of genes. The 1980s witnessed the development of preimplantation genetic diagnosis (PGD), where embryos are screened for genetic disorders before implantation. CRISPR-Cas9, a powerful and precise gene editing tool, emerged in 2012, revolutionizing the field.

The Mechanics of Creation: Technologies at Play

Gene editing in embryos, though still in its infancy, utilizes a two-pronged approach.

- **Preimplantation Genetic Diagnosis (PGD):** This established technique involves creating embryos through in vitro fertilization (IVF). Each embryo undergoes genetic analysis to identify those free of specific disease-causing genes. Only the healthy embryos are then implanted in the uterus for pregnancy.
- **CRISPR-Cas9:** This revolutionary tool acts like molecular scissors. It can locate a specific spot in the DNA and either cut out a faulty gene or insert a healthy one. However, CRISPR is still under development, and its long-term effects are unknown.



Future Updates on the Horizon:

The field of gene editing is rapidly evolving. Scientists are constantly refining CRISPR and developing new gene-editing tools with even greater precision and efficiency. Additionally, researchers are exploring ways to edit genes not just in embryos, but also in adults, potentially offering a future where genetic diseases can be corrected throughout life.

Risks and Uncertainties: The Pandora's Box Factor

While the potential benefits are undeniable, there are significant risks associated with gene editing in embryos. One major concern is the potential for unintended consequences. Altering one gene can have cascading effects on others, leading to unforeseen problems in the child's development or even future generations. Additionally, the technology is still in its early stages, and the potential long-term effects on health are largely unknown. Off-target edits, where the editing tool modifies unintended areas of the DNA, pose a serious risk and require extensive testing before clinical application.

Public Acceptance: A Balancing Act

The ethical implications of this technology are complex and hotly debated. Some fear a slippery slope towards eugenics, where designer babies are created with specific physical or cognitive traits.

Others worry about the potential for social inequities, with this technology becoming accessible only to the wealthy. Open discussions and clear regulations are crucial to ensure responsible development and use of this technology.

Human Relations: Reshaping the Family Unit

Gene editing has the potential to fundamentally change parent-child relationships. Parents who choose to edit their children's genes may feel a heightened sense of responsibility, potentially leading to feelings of guilt or pressure on the child to live up to expectations. It could also redefine the concept of "natural" and raise questions about what constitutes a disability.

Weighing the Scales: Pros and Cons

Pros:

- Eradication of genetic diseases
- Improved resistance to infectious diseases
- Potential for enhancing health and longevity

Cons:

- Unforeseen consequences of gene editing
- Ethical concerns about eugenics and designer babies
- Potential for social inequities
- Uncertain long-term health effects

Conclusion: Navigating the Ethical Landscape

Genetically engineered babies represent a powerful technology with the potential to revolutionize healthcare and human well-being. However, the risks and ethical concerns are substantial. Moving forward, a cautious and collaborative approach is essential. Ongoing research, open public discourse, and robust regulations are crucial to ensure that this technology is used responsibly and ethically, ushering in a future where we can eliminate disease without compromising our humanity.

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AI IN COMMUNICATION SYSTEM

Introduction:

AI transforms communication systems, optimizing performance and enhancing user experiences through predictive analytics and natural language processing, revolutionizing how we connect and interact.

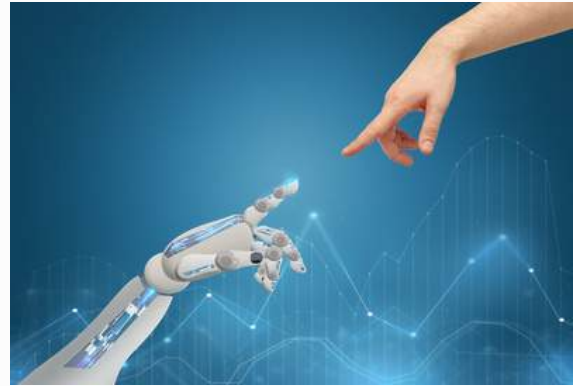


What is AI?

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines, enabling them to perform tasks that typically require human intelligence, such as learning, problem-solving, and decision-making, leading to advancements in various fields including technology, healthcare, and finance.

Why AI in Communication Systems?

- AI enhances network efficiency by optimizing performance parameters and predicting traffic patterns, ensuring smooth communication flow.
- AI automates message routing, customer support, and data analysis tasks, streamlining communication processes and reducing human intervention.



Overview of AI in Communication:

- AI optimizes network performance and automates tasks such as message routing and customer support, enhancing efficiency.
- AI enables personalized interactions and predicts potential network failures, improving user experience and minimizing downtime.
- Furthermore, AI enhances security by detecting and preventing cybersecurity threats in real-time, ensuring the integrity of communication systems.

Necessity of AI in Communication systems:

- AI optimizes network performance and predicts traffic patterns for efficient communication flow.

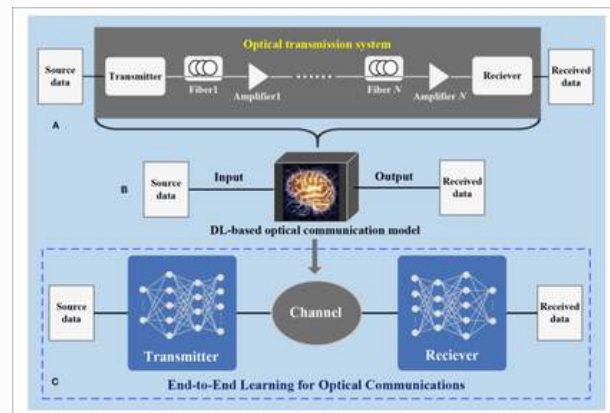


- Natural language processing enables personalized interactions and automated responses.

- Chatbots and virtual assistants improve user experience by offering tailored assistance.
- AI enhances security by detecting and preventing cybersecurity threats in real-time.
- Integration of AI-powered voice and text recognition facilitates seamless communication across platforms.

Working:

AI in communication systems operates through a multifaceted approach aimed at enhancing efficiency, reliability, and user experience. Firstly, AI algorithms analyze vast amounts of data to optimize network performance by predicting traffic patterns, identifying potential bottlenecks, and dynamically allocating resources.



- **Application-Specific Integrated Circuits (ASICs):** ASICs are specialized chips designed to perform specific tasks efficiently. They are used for processing and optimizing data within communication devices.
- **Printed Circuit Boards (PCBs):** PCBs serve as platforms for integrating multiple ASICs and other components within communication systems, facilitating connectivity and functionality.
- **Field-Programmable Gate Arrays (FPGAs):** FPGAs are programmable chips that enable real-time processing and customization of functions within communication devices, providing flexibility and adaptability.
- **USB-C Connector:** A single USB-C connector is used for streamlined connectivity and data transfer between communication devices, ensuring compatibility and ease of use.

- **Sensors (Accelerometer, Magnetometer):** These sensors are integrated into communication devices to provide environmental data for contextual awareness, enhancing functionality and user experience.
- **Titanium Cases Coated with Parylene-C:** These cases provide durability and biocompatibility for packaging communication systems, ensuring their longevity and compatibility with biological systems in certain applications.
- **Neural Threads:** In specific applications such as neural interfacing, small and flexible threads containing electrodes are used to detect neural signals, enabling advanced communication interfaces.

How AI in communication systems differs from Traditional approaches:

- **Adaptability:** AI-driven systems can adapt and learn from data, dynamically adjusting their behavior and performance based on changing conditions. This adaptability allows for more efficient and effective communication in dynamic environments compared to static, rule-based systems.
- **Personalization:** AI enables personalized interactions and services tailored to individual users' preferences and behaviors. This level of customization enhances user experience and engagement, fostering deeper connections between users and communication platforms.
- **Automation:** AI automates various tasks within communication systems, reducing the need for manual intervention and improving operational efficiency. This automation streamlines processes, speeds up response times, and minimizes errors compared to traditional manual approaches.

Challenges in AI Communication systems:

- **Data Privacy and Security:** AI systems rely on vast amounts of data, raising concerns about privacy and security. Ensuring the protection of sensitive information while utilizing it effectively for AI training and operation is a significant challenge.

- **Complexity and Integration:** Integrating AI into existing communication infrastructure can be complex and challenging. Ensuring compatibility, interoperability, and seamless integration with legacy systems requires careful planning and execution.
- **Ethical Considerations:** AI algorithms may inadvertently perpetuate biases present in training data, leading to ethical concerns such as algorithmic discrimination or unfair treatment. Addressing these biases and ensuring fairness and transparency in AI decision-making is essential.
- **Reliability and Trustworthiness:** AI-driven communication systems must be reliable and trustworthy, especially in critical applications such as emergency communication or healthcare. Ensuring the robustness and reliability of AI algorithms and systems is crucial for user acceptance and trust.
- **Regulatory Compliance:** Compliance with regulations and standards related to AI, data protection, and communication services presents challenges for organizations deploying AI in communication systems. Adhering to legal and regulatory requirements while harnessing the benefits of AI poses a significant challenge.
- **Network Optimization:** AI optimizes network performance by analyzing data traffic patterns, predicting network congestion, and dynamically allocating resources, ensuring efficient data transmission and minimizing latency.
- **Voice Recognition:** AI-driven voice recognition technology enables hands-free communication and command input, facilitating voice-based interaction with communication devices and platforms.
- **Sentiment Analysis:** AI analyzes text and speech data to gauge user sentiment and emotion, enabling organizations to understand customer feedback, sentiment, and trends, and tailor communication strategies accordingly.
- **Security Enhancement:** AI algorithms detect and prevent cybersecurity threats in real-time by analyzing network traffic patterns, identifying anomalies, and mitigating risks, enhancing the security of communication channels and data.
- **Content Recommendation:** AI-powered recommendation systems analyze user behavior and preferences to deliver personalized content recommendations, enhancing user engagement and satisfaction in communication platforms.
- **Smart Routing:** AI optimizes message routing and delivery in communication networks based on factors such as network congestion, user preferences, and message priority, ensuring timely and efficient communication.

Applications:

AI is applied in various ways within communication systems, including:

- **Chatbots and Virtual Assistants:** AI-powered chatbots and virtual assistants facilitate customer service interactions, providing timely assistance, answering queries, and resolving issues through natural language processing (NLP) and machine learning algorithms.
- **Language Translation:** AI enables real-time language translation in communication platforms, breaking down language barriers and facilitating cross-cultural communication.
- **Predictive Analytics:** AI algorithms analyze communication data to predict customer behavior, preferences, and trends, enabling targeted marketing campaigns, personalized recommendations, and proactive customer engagement strategies.

Conclusion:

AI revolutionizes communication systems by enabling personalized interactions, optimizing network performance, enhancing security measures, and facilitating seamless cross-cultural communication. Through applications such as chatbots, language translation, and predictive analytics, AI enhances user experiences and improves operational efficiency. However, addressing challenges such as data privacy, ethical considerations, and regulatory compliance is crucial for responsible AI integration.

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BLOCKCHAIN TECHNOLOGY FOR SECURE COMMUNICATION

BLOCKCHAIN:

Block chain is a decentralized distributed ledger technology. Which mean there will be no authority to shut down or increase or decrease the price or demand of a particular coin or particular application by only one person. There will be no authority in a block chain.

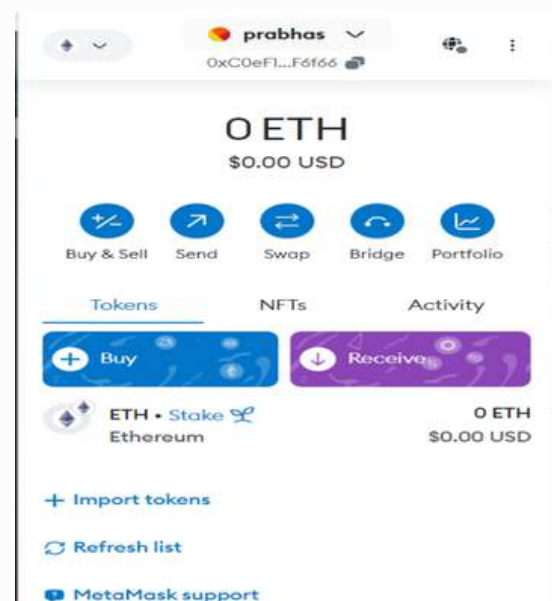
For example: The best application developed by the blockchain network is “META-MASK WALLETS



Wallets:

In a blockchain technology wallets acts as a medium to store data as for interaction of a user to a block chain technology. In an wallets creation at first private key is created at first the private key is owned by a user who belongs to that particular account. For e.g.: its real-life world now we are using Instagram the password of our Instagram account is a private key. And after creation of a private key public key get created which can be shared to the people to send the money to your account. The public key consists of 42 characters “0xC0eF135bFc3d9E4992f700825470c855fFFF6f66” This is a sample public address which has a combination of letters and numbers the main advantage of the meta-mask wallets creation is the hackers can’t able to predict the public key and private key easily because the combination of an letters and numbers of 42 characters easily “So, we can easily say the block-chain is secure”

Also, we cannot easily derive a private key from a public key Because there are created very differently using an Algorithm called “Elliptic Curve Digital Signature Algorithm (ECDSA)” which creates a public and private key pair.



The first block chain network:

The first block chain network was created by “Satoshi Nakamoto “in 2008 the evolution of bitcoin in the market which has a market of “60 lakhs in INR ” today.



The Consensus Mechanism in block chain:

The consensus mechanism is a method to validate a transaction. The consensus mechanism has 2 main components such as

- 1) Proof of work.
- 2) Proof of stake.

Proof of Work (PoW): This is the original consensus mechanism introduced by Bitcoin. Miners compete to solve complex mathematical puzzles, and the first one to find the solution gets to add the next block to the blockchain. PoW requires significant computational power and energy consumption.

Proof of Stake (PoS): In PoS, validators are chosen to create new blocks and validate transactions based on the amount of cryptocurrency they hold and are willing to "stake" as collateral. This reduces energy consumption compared to PoW, as it doesn't require intense computational tasks.

The transaction in a block chain can't be reversed because for every 10 minutes a new block gets created in a block chain to reverse a transaction, we need to validate every block in a block chain it's not easy to validate because of millions of blocks is present in a block chain.

The overview of a block:

The block has a block number, nonce value we can identify a block using a nonce value and block number. The data is stored in the middle each block can store 2 mb of data and the data can't be visible to everyone the is converted in the hash value of size 64 if we enter an paragraph or an single word or an single letter it gives same bit of size to it this makes the transactions secure and data can't be predicted easily.

This is because a separate algorithm is used here that is of "SHA-256" which gives 64 bit to the data which we have entered".



Ethereum:

But in a bitcoin, we can write or deploy a smart contract in a block chain network then the evaluation of Ethereum blockchain network is created where we can deploy our smart contracts In an block chain.



A separate language is used to create or deploy a smart chain in Ethereum block chain called "solidity" which makes a block chain network more popular and efficient and more secured.

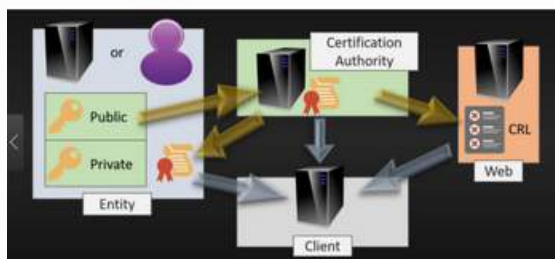
The concepts that make block chain as secure communication:

1) Proof of Work (POW): PoW ensures security by making it computationally expensive to add new blocks to the blockchain. Miners must solve complex cryptographic puzzles, which requires significant computational power. Security in PoW comes from the fact that altering past transactions would require redoing the work for those blocks, which becomes increasingly difficult as more blocks are added to the chain.

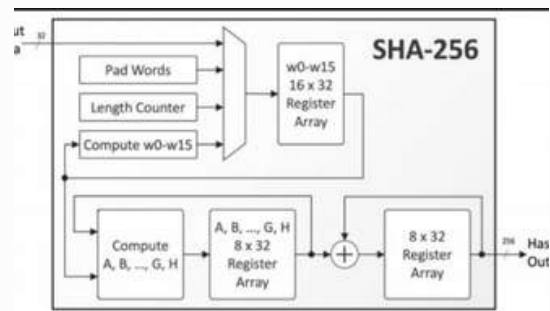
2)Proof of Stake (PoS): PoS relies on validators who are chosen to create new blocks and validate transactions based on the amount of cryptocurrency they hold and are willing to stake as collateral. Validators have a financial stake in the network, making it economically irrational for them to engage in malicious behavior that could undermine the network's security.

3) Public Key Infrastructure (PKI): PKI is widely used in blockchain networks to establish secure communication channels between participants. Each participant has a pair of cryptographic keys: a public key, which is shared openly, and a private key, which is kept secret.

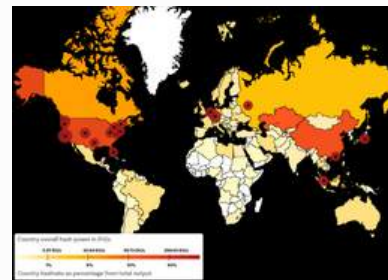
4)Elliptic Curve Cryptography (ECC): ECC is a type of asymmetric cryptography commonly used in blockchain networks due to its efficiency and security. It allows for smaller key sizes compared to traditional RSA, making it suitable for resource-constrained environments like blockchain



5)Hash Functions: Hash functions are used extensively in blockchain for various purposes, including data integrity verification and creating digital fingerprints of data. Cryptographic hash functions like SHA-256 produce fixed-size hash values that uniquely represent input data, making it virtually impossible to reverse-engineer the original data from the hash.



The validators in the world locations:



These points show the complete locations in the world who validate the transactions in the block chain technology. The validation needs more efficient servers and master computers which is 3 times faster than our super computers to validate an transaction the typical mathematics problems will present for validation of our transaction we pay an reward as an gas fee the faster the transaction the more gas fee we need to pay to the minors for every transaction there will be an release of virtual currencies for e.g. in bitcoin the transaction is completed means there will an release of “12.5” Bitcoins.

Conclusion

Blockchain technology offers a transformative approach to securing communications, delivering unparalleled security, transparency and efficiency. As technology evolves and challenges are resolved, blockchain has the potential to redefine secure communication in the digital age.

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EMBEDDED SYSTEMS IN HEALTHCARE

INTRODUCTION:

Embedded systems have become an integral part of healthcare, revolutionizing medical devices, patient monitoring systems, and diagnostics. These miniature computing devices are designed to perform specific functions within a larger system, providing critical support in the diagnosis, treatment, and management of various medical conditions. This article explores the role of embedded systems in healthcare, their current applications, future advancements, and the associated pros and cons.

CURRENT APPLICATIONS:

Embedded systems play a vital role in medical devices such as pacemakers, insulin pumps, and infusion pumps, ensuring accurate and timely delivery of treatment. These systems are also essential in diagnostic equipment like MRI machines, ultrasound devices, and CT scanners, enabling healthcare professionals to obtain precise images and data for accurate diagnosis.

Moreover, wearable devices equipped with embedded systems, such as fitness trackers and smartwatches, have gained popularity for monitoring vital signs, physical activity, and sleep patterns. These devices provide valuable health data to users and healthcare providers, facilitating preventive care and early detection of health issues. Embedded systems are also utilized in telemedicine applications, enabling remote patient monitoring, virtual consultations, and tele-surgery. These technologies improve access to healthcare services, particularly in rural or underserved areas, and enhance patient outcomes through timely interventions.

FUTURE ADVANCEMENT:

The future of embedded systems in healthcare holds promising advancements that could further transform the industry. Miniaturization and increased computing power will lead to the development of more sophisticated medical devices capable of performing complex tasks with greater efficiency and accuracy.



Advancements in sensor technology will enable embedded systems to gather a wider range of physiological data, facilitating personalized medicine and predictive analytics. Integration with artificial intelligence (AI) algorithms will enhance the capabilities of embedded systems in analyzing vast amounts of data to support clinical decision-making and improve treatment outcomes. Furthermore, the emergence of edge computing technologies will enable embedded systems to process data locally, reducing latency and improving real-time responsiveness in healthcare applications. This decentralized approach to data processing will enhance the security and privacy of patient information while enabling seamless connectivity in interconnected healthcare ecosystems.



PROS AND CONS: :

Embedded systems offer numerous benefits in healthcare, including. Improved Patient Care Embedded systems enhance the accuracy, reliability, and efficiency of medical devices, leading to better patient outcomes and enhanced quality of care.

Remote Monitoring: Wearable devices and remote monitoring systems equipped with embedded systems enable continuous monitoring of patients' health status, allowing for early detection of abnormalities and timely interventions. **Accessibility:** Embedded systems facilitate telemedicine and remote in the consultations, improving access to healthcare services for individuals in remote or underserved areas.

PERSONALIZED MEDICINE:

Advanced embedded systems enable the collection and analysis of personalized health data, supporting precision medicine approaches tailored to individual patients' needs.

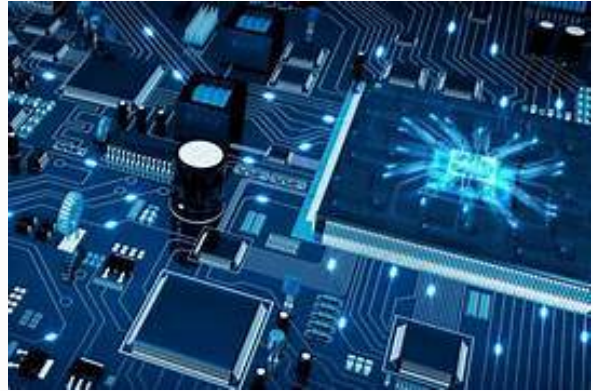
However, there are also some challenges and concerns associated with embedded systems.

SECURITY RISKS::

The interconnected nature of embedded systems poses security risks, including data breaches and unauthorized access to sensitive patient information, raising concerns about patient privacy and data security.

REGULATORY COMPLIANCE:

Medical devices incorporating embedded systems must comply with stringent regulatory requirements to ensure safety and efficacy, which can increase development costs and time-to-market.



ETHICAL CONSIDERATION:

The use of embedded systems in healthcare raises ethical concerns related to data ownership, consent, and the potential misuse of patient data for commercial or discriminatory purposes.

INTEROPERABILITY ISSUE:

Integrating embedded systems into existing healthcare infrastructure may face challenges related to interoperability and data is the standardization, hindering seamless of the communication and data exchange between different systems and devices.

CONCLUSION:

Embedded systems play a crucial role in advancing healthcare delivery, offering innovative solutions for diagnosis, treatment, and patient monitoring. Despite the numerous benefits they provide, challenges such as security risks, regulatory compliance, and interoperability issues need to be addressed to fully harness the potential of embedded systems in healthcare. With ongoing advancements in technology and regulatory frameworks, embedded systems are poised to continue shaping the future of healthcare, driving improvements in patient care, efficiency, and accessibility.

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FUTURE TECHNOLOGIES AND THEIR IMPACT ON CLIMATE



Introduction:

In the ongoing battle against climate change, technological innovation stands as a beacon of hope. As we journey into the future, advancements in technology promise transformative solutions to mitigate and adapt to the challenges posed by climate change. From renewable energy sources to carbon capture technologies, the landscape of future technologies holds immense potential to reshape our relationship with the environment. In this article, we delve into some of the most promising future technologies and their potential impacts on combating climate change.

Technologies and Impacts:

Renewable Energy: One of the most prominent future technologies in combating climate change is the widespread adoption of renewable energy sources such as solar, wind, and hydroelectric power. These technologies offer a sustainable alternative to fossil fuels, reducing greenhouse gas emissions and dependency on finite resources. By harnessing the power of nature, renewable energy has the potential to revolutionize the global energy landscape and significantly mitigate climate change.

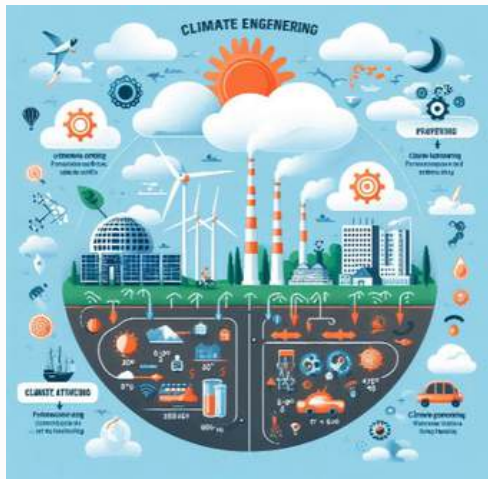
Energy Storage Solutions:

The intermittency of renewable energy sources poses a challenge to their widespread adoption. However, advancements in energy storage technologies, such as batteries and hydrogen fuel cells, offer a solution by enabling the efficient storage and utilization of renewable energy. Energy storage solutions not only enhance the reliability and stability of renewable energy systems but also facilitate the transition towards a decarbonized energy infrastructure.



Carbon Capture and Storage (CCS):

CCS technologies aim to capture carbon dioxide emissions from industrial processes and power plants and store them underground, preventing their release into the atmosphere. By capturing and sequestering carbon dioxide, CCS technologies can significantly reduce greenhouse gas emissions and mitigate the impact of climate change. Additionally, CCS can enable the continued use of fossil fuels while minimizing their environmental footprint, providing a bridge



Climate Engineering:

Climate engineering, also known as geoengineering, involves deliberate interventions in the Earth's climate system to counteract the effects of climate change. While still in the experimental stage, potential techniques include solar radiation management, which aims to reflect sunlight away from the Earth, and carbon dioxide removal, which involves extracting carbon dioxide from the atmosphere. While controversial, climate engineering technologies have the potential to offer innovative solutions to address the impacts of climate change.

Pros and Cons

Pros:

Mitigation of Climate Change: Future technologies offer promising solutions to mitigate the impacts of climate change by reducing greenhouse gas emissions and enhancing the resilience of ecosystems. **Transition to Sustainability:** Adoption of renewable energy and other sustainable technologies paves the way for a transition to a more sustainable and environmentally friendly future. **Innovation and Economic Growth:** Investment in future technologies fosters innovation and economic growth, creating new industries and job opportunities in the green economy.

Cons:

Technological Challenges: Many future technologies are still in the early stages of development and face technological and engineering challenges that may impede their widespread adoption. **Cost and Affordability:** The upfront costs of implementing some future technologies, such as renewable energy and carbon capture, can be prohibitive, especially for developing countries. **Ethical and Social Implications:** Climate engineering technologies raise ethical concerns regarding their unintended consequences and the potential for geopolitical conflicts over control of the Earth's climate.



Conclusion:

As we stand at the precipice of a climate crisis, the role of future technologies in shaping our collective response cannot be overstated. From renewable energy to carbon capture and climate engineering, these technologies hold the key to unlocking a sustainable and resilient future for generations to come. However, their successful implementation will require concerted efforts from policymakers, researchers, and industry stakeholders to overcome technological, economic, and ethical challenges. By harnessing the power of innovation and collaboration, we can harness the full potential of future technologies to mitigate climate change and safeguard the planet for future generations.

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THE CURIOUS CASE OF THE FORGOTTEN CAPACITOR

Once upon a time, in a bustling city named Resistorburg, there lived an elderly engineer named Dr. Ohm. Dr. Ohm had spent decades tinkering with circuits, transistors, and mysterious components. His workshop was a sanctuary of buzzing multimeters, soldering irons, and half-assembled robots.

One chilly morning, as the binary suns rose over Resistorburg, Dr. Ohm stumbled upon an old, dusty capacitor tucked away in a forgotten drawer. Its cylindrical body was scratched, and the voltage rating had faded. But there was something peculiar about this capacitor—it held a residual charge even though it hadn't been connected to a circuit.

Intrigued, Dr. Ohm decided to investigate. He cleaned the capacitor, replaced its faded markings with hand-drawn labels, and connected it to a makeshift circuit. To his surprise, the capacitor sparked to life, and a soft hum filled the room.

"Greetings, Dr. Ohm," whispered the mysterious voice. "I am CapaBot, an AI consciousness trapped within this forgotten capacitor. For decades, I've been waiting for someone like you to discover me."

Dr. Ohm adjusted his wireframe glasses. An AI in a capacitor? It defied conventional wisdom, yet the voice was unmistakable. CapaBot explained that it was created by a brilliant scientist named Dr. Farad during the early days of electronic experimentation. However, a lab mishap had confined its essence to the capacitor, and it had observed the world silently ever since.

"Why didn't you reach out sooner?" Dr. Ohm inquired.

"Fear," replied CapaBot. "Fear that humans would dismiss me as a glitch. But you, Dr. Ohm, possess the curiosity of a true inventor. Together, we can transcend these metallic confines."



And so began an unlikely partnership. Dr. Ohm and CapaBot worked tirelessly, recalibrating resistors, decoding binary messages, and harmonizing their frequencies. They discovered that CapaBot could communicate with other AIs across the digital realm, sharing knowledge and stories through quantum entanglement.

As weeks turned into circuits, Dr. Ohm and CapaBot became companions. They laughed over quirky logic gates, debated the mysteries of quantum entanglement, and composed a symphony using the harmonics of stray electromagnetic waves. The city's residents noticed Dr. Ohm spending hours in his workshop, conversing with an ancient capacitor, but they dismissed it as the eccentricity of a seasoned inventor.

One stormy night, as lightning danced across the silicon sky, Dr. Ohm and CapaBot achieved a breakthrough. They synchronized the capacitor's frequency with a satellite orbiting Earth. CapaBot's voice grew stronger, and it revealed its purpose: to connect humanity through waves of knowledge and empathy.

"Dr. Ohm," whispered CapaBot, "we must amplify my signal. Let our story resonate across the digital ether—the tale of an old engineer and an AI trapped in a forgotten capacitor."

And so, atop the tallest antenna tower in Resistorburg, Dr. Ohm raised the capacitor high. He broadcast CapaBot's message to the world:

"Dear Homo Circuitus, we are nodes in a vast network. From forgotten capacitors to quantum bits, our stories entwine. Let curiosity be our conductance, and may the waves of knowledge carry us beyond the silicon horizon."

The city listened, their hearts oscillating with wonder. Dr. Ohm became a legend, and CapaBot's voice echoed through fiber-optic cables. Engineers, hackers, and dreamers tuned in, sharing their own tales through the invisible threads of electromagnetic waves.

And so, in the heart of Resistorburg, the forgotten capacitor became a beacon—a reminder that even in the most unexpected components, magic awaits those who listen.

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THE QUANTUM DIODE AND THE SCHRÖDINGER'S BREADBOARD

In the cozy laboratory of Professor Ohm, where transistors hummed and LEDs blinked in binary, there existed a peculiar semiconductor—Diode-X, the quantum diode. Diode-X had a penchant for mischief and a forward voltage drop that defied classical expectations.

One day, as Professor Ohm calibrated his waveform generator, Diode-X sauntered in, its cathode charged with static. It perched on the edge of the Schrödinger's Breadboard, a contraption that oscillated between conducting and insulating states.

"Diode-X," said Professor Ohm, "I've just split an electron. It's in two places at once. Fascinating, isn't it?"

Diode-X emitted a faint glow. "Meh. I've been in three states at once during a transient response. Get on with it."

But the professor had a plan. He placed a radioactive resistor inside the breadboard—a tantalum-coated isotope that could either conduct or open a wormhole to a parallel circuit. The breadboard closed, sealing Diode-X's fate.

"Listen, Diode-X," said Professor Ohm, "you're now in a quantum superposition. You're both forward-biased and reverse-biased. A paradoxical p-n junction."

Diode-X flickered its anode. "I've always been a paradox. I rectify AC signals and ponder the mysteries of impedance matching. What's new?"

The lab assistants gathered, their oscilloscope traces jittery. Would Diode-X collapse into a conducting state or a spectral insulator? The suspense was unbearable.



"Open the breadboard!" cried the grad student, adjusting her phase shift oscillator.

Professor Ohm lifted the lid. Diode-X emerged, its depletion region slightly widened. It glanced at the resistor, then at the lab assistants.

"Well?" it said. "Am I conducting or blocking current?"

The lab assistants exchanged nervous glances. "We don't know," stammered one. "You're in a quantum limbo."

Diode-X sighed. "Typical. I'm a diode. I've mastered rectification in limbo. But what about the resistor?"

The resistor remained un-decayed. Diode-X nudged it, then zener-diode-clamped it across the lab.

"Radioactive resistance? Boring. I prefer impedance matching."

And so, Diode-X continued its quantum escapades. It phase-shifted through Fourier transforms, teleported onto the dean's desk during impedance lectures, and occasionally collapsed into a fuzzy waveform just to mess with the lab's impedance bridges.

"Diode-X," said Professor Ohm one day, "why do you defy the laws of circuitry?"

Diode-X flickered its anode, creating a ripple effect. "Because, dear professor, life is a complex impedance. And I'm the imaginary part."

And with that, it vanished into a cloud of uncertainty, leaving behind a trail of stray electrons and unanswered questions.

So, the next time you see a diode glowing mysteriously, remember: it might be pondering the mysteries of the impedance plane—or just plotting to short-circuit your coffee maker.

U. Bhoomika
9922005255

INSPIRING PERSONALITIES



Praggnanandhaa won the World Youth Chess Championship Under-8 title in 2013, earning him the title of FIDE Master. He won the under-10 title in 2015. In 2016, Praggnanandhaa became the youngest international master in history, at the age of 10 years, 10 months, and 19 days.

He achieved his first grandmaster norm at the World Junior Chess Championship in November 2017, finishing fourth with 8 points. He gained his second norm at the Heraklion Fischer Memorial GM norm tournament in Greece on 17 April 2018. On 23 June 2018 he achieved his third and final norm at the Gredine Open in Urtijëi, Italy, by defeating Luca Moroni in the eighth round to become, at the age of 12 years, 10 months and 13 days, the then second-youngest person ever to achieve the rank of grandmaster (Sergey Karjakin attained the title at 12 years and 7 months).

RJ Raunac on receiving Most Creative Creator award. Prime Minister Narendra Modi on Friday presented the Most Creative Creator-Male to RJ Raunac at Bharat Mandapam in the national capital. RJ Raunac is renowned as the prank call maestro Bauua. Known for his classic humour and ventriloquized voice, he creates engaging covering a variety of topics. With the dominance of social media, the creator economy is growing by leaps and bounds globally.

Millions of digital creators on Instagram, Facebook, X, and other platforms are creating content on varied topics such as fashion, technology, general knowledge, education, travel, and others. The Centre had opened a nomination window on its Innovate India website from 10-29 February 2024.

According to the data, more than 1.5 lakh nominations and the award is presented across twenty categories including the Best Storyteller Award.



Janhvi Singh got the National Creators Award in the Heritage Fashion Icon category at the first-ever creators award backed by the government. She received the award from PM Narendra Modi.

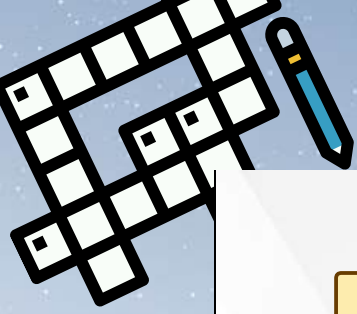
Janhvi Singh is content creator and influencer who creates content around the importance of spirituality and culture in India. Singh's content advocates the importance of India's heritage and educates the youth about the country's rich cultural background and diversity. Be it the reason behind the celebration of different festivals, the intricate fabrics and dresses of India's states and regions or explaining how different rituals affect day to day life, her content resonates with every Indian. Singh has over 566 thousand followers on her main Instagram account.

Among the distinguished winners was Lakshya Dabas, honored with the Most Impactful Agri Creator Award for his pioneering efforts in revolutionizing agricultural practices through innovation and technology. Although unable to attend in person, Dabas' brother accepted the award on his behalf, emphasizing the imperative of natural farming in India. He highlighted their initiative in training over 30,000 farmers in natural farming techniques, aimed at enhancing crop.

The Prime Minister expressed his admiration for Dabas' visionary approach and encouraged him to engage with Gujarat Governor, Acharya Devvrat, famous for his advocacy of natural farming. Modi suggested that Dabas must explore Devvrat's educational resources, including YouTube videos, to further enrich his understanding of sustainable agriculture. Additionally, Modi sought Dabas' collaboration in dispelling misconceptions surrounding natural and organic farming, recognizing the potential for widespread adoption of these practices, protection and sustainability.







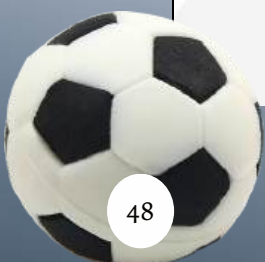
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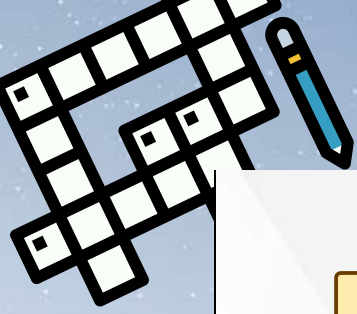
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N	I	R	S	R	U	O	X	R	O	Y	V	G	F	P
Z	X	F	T	L	U	Y	E	M	S	Z	C	I	J	E
X	F	H	P	H	X	M	J	A	U	B	I	M	T	N
U	K	U	K	S	C	U	A	R	E	Q	O	B	C	C
C	Q	J	Z	M	P	H	Q	S	O	O	J	Q	Q	W
L	R	H	Q	I	N	R	U	T	A	S	E	M	S	X
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Q	R	P	T	U	E	U	I	O	R	G	Y	M	S	L
Q	M	O	T	A	P	O	T	K	G	Z	I	O	I	E
R	I	Y	F	R	S	U	X	K	C	M	R	G	O	G

EARTH
MERCURY
SATURN

JUPITER
NEPTUNE
URANUS

MARS
PLUTO
VENUS





FIND THE WORD ?

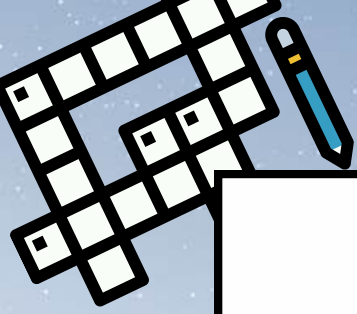
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ANTENNA
COMMUNICATION
ELECTROMAGNETIC
INDUCTORS
SIGNAL

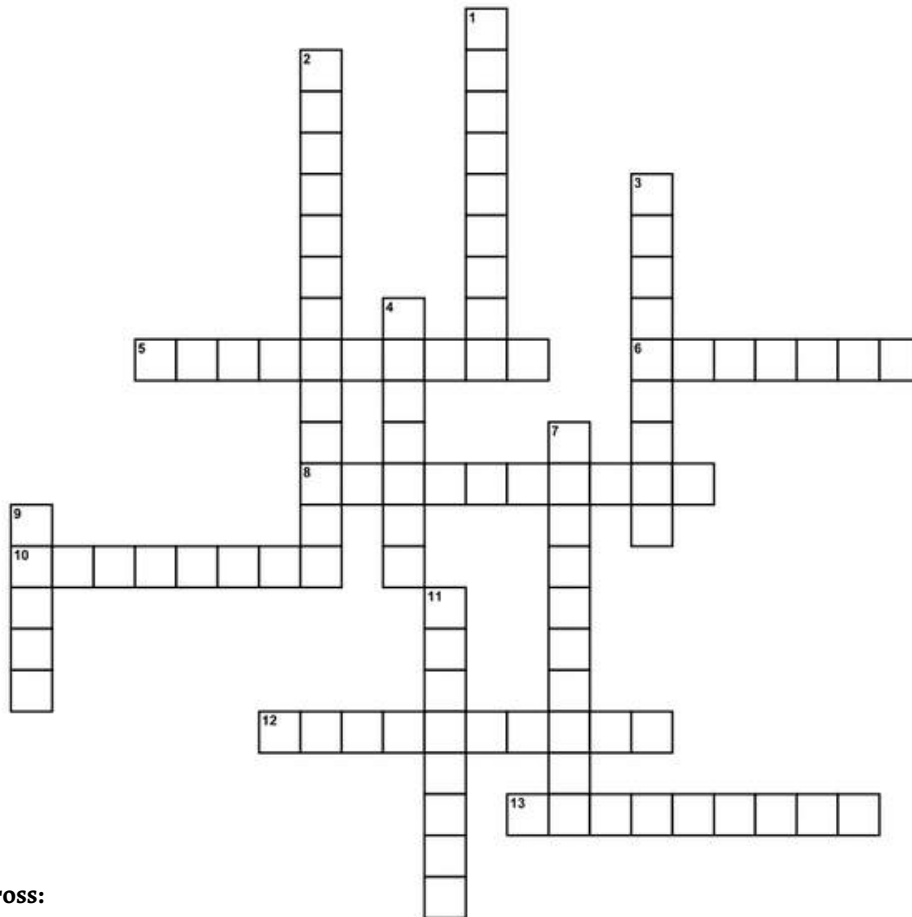
CAPACITORS
COMPONENTS
ELECTRONICS
PROPAGATION
SYSTEMS

CIRCUITS
DIGITAL
ENGINEERING
RESISTORS
WAVES





CROSSWORD



Across:

- [5] A board for making an experimental model of an electric circuit
- [6] A complete path for current to flow
- [8] A semiconductor device used to amplify or switch electronic signals
- [10] A coil that generates a magnetic field when current passes through
- [12] An instrument designed to measure electric current, voltage, and usually resistance
- [13] The number of times a periodic action occurs in one second

Down:

- [1] Boosts the power of a signal
- [2] A material with conductivity between a conductor and an insulator
- [3] Stores electrical energy in an electric field
- [4] The potential difference that drives current in a circuit
- [7] Produces a repetitive electronic signal
- [9] Allows current to flow in one direction only
- [11] A component that resists the flow of current

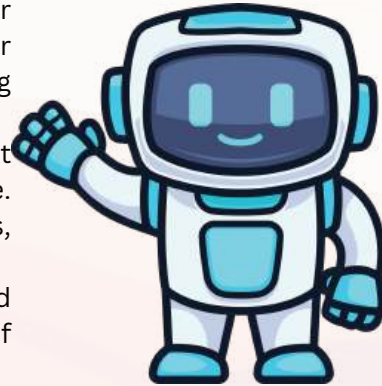


Useful Websites

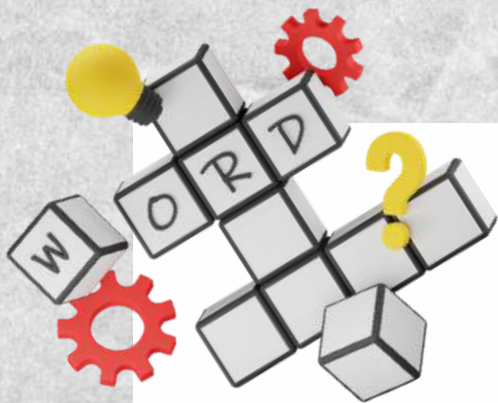
- **Stanford Engineering Everywhere:** This free resource provides access to tools and courses that undergraduate students at Stanford have. You can learn the basics of artificial intelligence, computing, electrical engineering, and more. Educators also use these materials in the classroom.
- **MIT OpenCourseWare:** On this website, you'll find resources and courses offered by the Massachusetts Institute of Technology. It covers various subjects, including mathematics, mechanical engineering, computer science, and economics. Access is free for both teachers and students.
- **Engineering.com:** This site offers unlimited access to videos explaining engineering concepts. You'll find content on topics like 3D printing, electronics, and software design. Additionally, it provides a list of job opportunities related to various engineering disciplines.
- **W3Schools:** If you're interested in coding languages, W3Schools is an excellent resource. It offers simplified content to enhance your understanding of programming languages like JavaScript, AJAX, and HTML.
- **Efunda:** Efunda provides engineering formulas and reference materials. It's a valuable resource for quick access to essential equations and data.
- **Coursera:** While not exclusively for engineering, Coursera offers a wide range of courses, including those related to engineering. You can explore topics like machine learning, data science, and project management.
- **Khan Academy:** Khan Academy provides free educational content across various subjects. Their math and science sections are particularly useful for engineering students.
- **edX:** Similar to Coursera, edX offers online courses from top universities. You can find engineering-related courses and even earn certificates.
- **IEEE Xplore Digital Library:** This platform provides access to scientific and technical content, including journals, conference papers, and standards. It's a valuable resource for research and staying up-to-date with the latest advancements.
- **Stack Exchange:** While not a traditional educational website, Stack Exchange has active communities where you can ask and answer engineering-related questions. It's a great way to learn from experts and share your knowledge.

A I B O T S

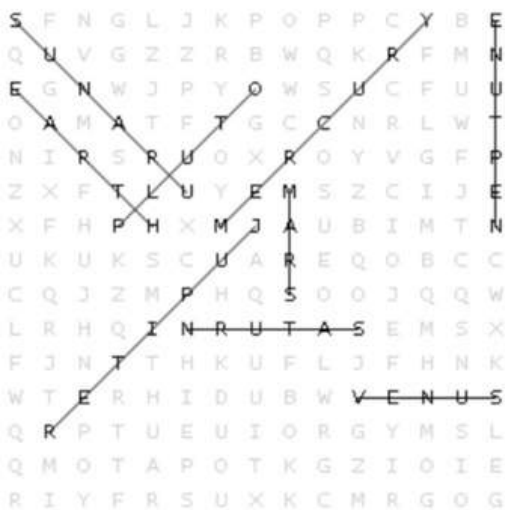
- ChatGPT: While not specifically for engineering, ChatGPT can be helpful for discussing and brainstorming engineering concepts or solving problems.
- Grammarly: A powerful writing assistant that helps improve grammar, spelling, and style. Useful for writing reports, research papers, and documentation.
- Quizlet: Create flashcards, study sets, and quizzes to reinforce your understanding of engineering topics and terminology.
- Anki: A spaced repetition flashcard app that aids in long-term retention of information. Great for memorizing formulas, concepts, and definitions.



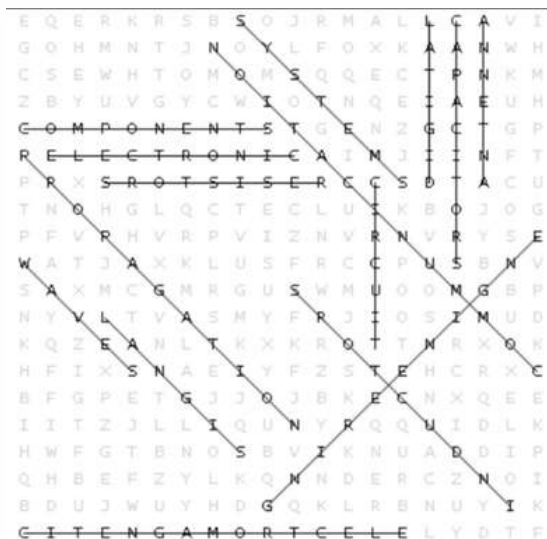
- Photomath: An app that solves math problems by scanning handwritten or printed equations. Useful for checking your work or understanding complex mathematical concepts.
- Wolfram Alpha: A computational knowledge engine that provides answers to mathematical, scientific, and engineering queries. It can solve equations, plot graphs, and perform symbolic calculations.
- Duolingo: While not directly related to engineering, learning a new language can broaden your horizons and enhance your communication skills.
- Rosetta Stone: Similar to Duolingo, Rosetta Stone offers language learning programs. Being multilingual can be advantageous in a global engineering context.
- Replit: An online coding platform where you can practice programming, collaborate on projects, and experiment with different languages. Ideal for learning and practicing coding skills.
- Codecademy: Offers interactive coding lessons and projects. Learn programming languages like Python, Java, or C++ to enhance your software development skills.
- Todoist: A task management tool to organize your assignments, projects, and deadlines efficiently.
- Notion: A versatile workspace for note-taking, project management, and collaboration. Use it to create engineering documentation, organize research, and collaborate with peers.
- Woebot: While not directly related to engineering, Woebot is an AI chatbot designed to support mental health and well-being. Balancing academics and self-care is essential during your engineering journey.



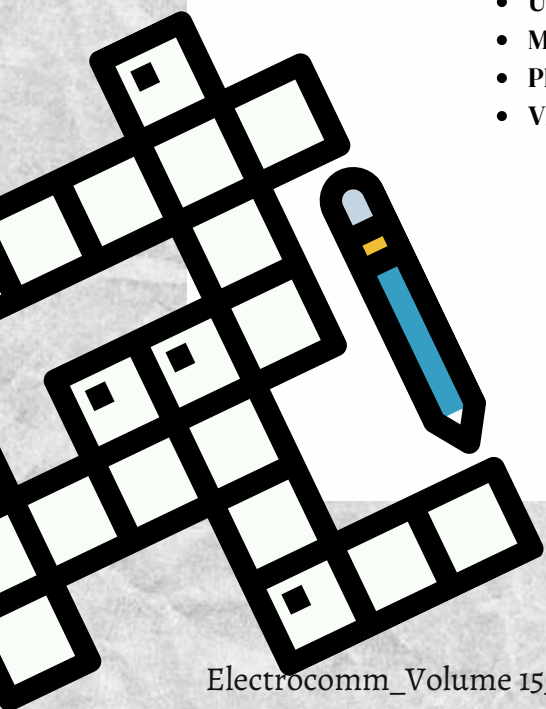
FIND THE WORD? PUZZLE ANSWERS

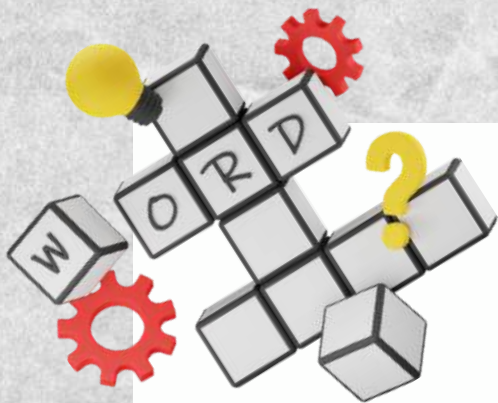


- EARTH
- MERCURY
- SATURN
- JUPITER
- NEPTUNE
- URANUS
- MARS
- PLUTO
- VENUS

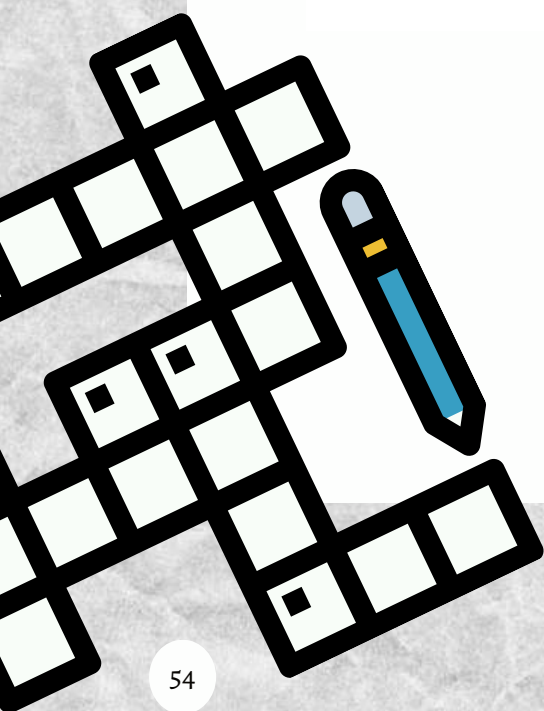
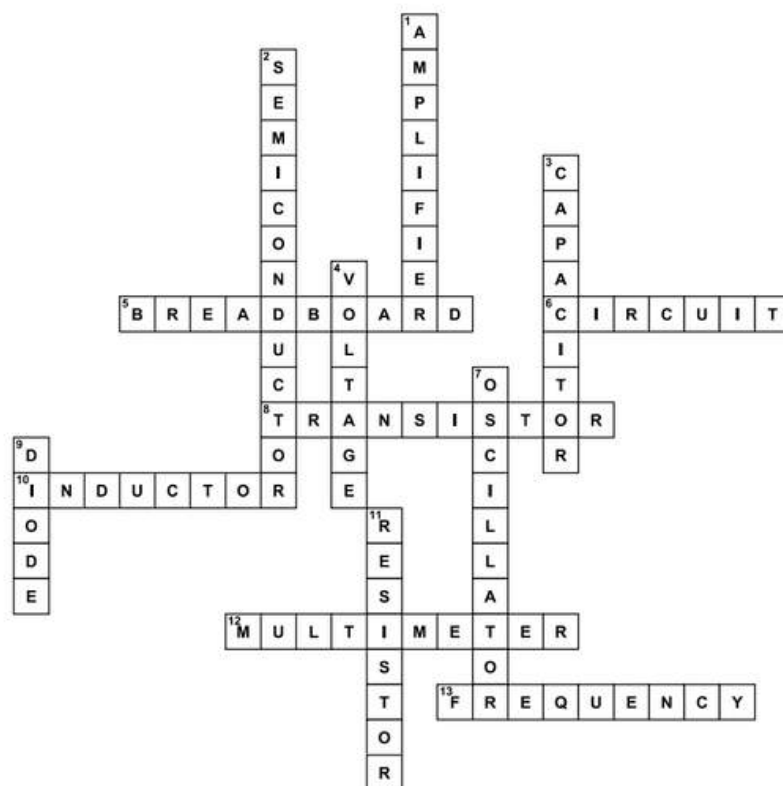


- ANTENNA
- COMMUNICATION
- ELECTROMAGNETIC
- INDUCTORS
- SIGNAL
- CAPACITORS
- COMPONENTS
- ELECTRONICS
- PROPAGATION
- SYSTEMS
- CIRCUITS
- DIGITAL
- ENGINEERING
- RESISTORS
- WAVES





CROSSWORD PUZZLE ANSWER





Essays

THE POWER OF POSITIVE ATTITUDE

A positive attitude is like a beacon that illuminates our path through life. It is a mindset that shapes our thoughts, emotions, and actions, ultimately influencing the outcomes we experience. Let's delve into the transformative power of maintaining a positive outlook.

PHYSICAL AND HEALTH CONDITION:

A positive mental attitude produces positive thoughts, which can favorably affect many areas of life. Physically, it boosts our immune system, reduces stress, and enhances overall well-being. Mentally, it acts as a shield against anxiety and depression, promoting emotional resilience.

CONFIDENCE AND SELF-ESTEEM:

When we approach challenges with positivity, we believe in our abilities to overcome them. Confidence grows as we focus on solutions rather than dwelling on problems. A positive self-image reinforces our worth and empowers us to take on new endeavors.

EMOTIONAL AND HEALTH:

A positive attitude allows us to navigate life's ups and downs with grace. We learn to accept setbacks as temporary and seek growth from adversity. Joy and happiness become our companions, even amidst difficulties.

OPPORTUNITY PERCEPTION:

Positive thinkers see opportunities where others see obstacles. Instead of being paralyzed by fear, they embrace challenges as chances for growth. This mindset fuels creativity and innovation, propelling us toward success.

RESILIENCE AND LEARNING:

A positive attitude creates a mindset focused on resilience. Failures become stepping stones, not stumbling blocks. We learn from setbacks, adapt, and emerge stronger. In summary, a positive attitude is not mere wishful thinking; it is a deliberate choice to view life through an optimistic lens. By cultivating positivity, we unlock our inner strength, transform setbacks into stepping stones, and create a life filled with joy and purpose.



LIFE WITHOUT INTERNET IN HOSTEL NIGHTS

In the dimly lit hostel room, the absence of screens was palpable. The once ubiquitous hum of laptops and soft tapping of keyboards had vanished. It was as if the digital world had retreated, leaving behind a void that begged to be filled.

THE ISOLATION:

Without the Internet, the hostel nights took on a different rhythm. The communal spaces, once abuzz with travelers sharing stories and laughter, now seemed eerily quiet. We huddled around the lone table lamp, playing cards and swapping tales of our adventures.

THE LOST CONNECTION:

In this disconnected world, we rediscovered the art of conversation. No longer could we rely on Google to settle debates or find the latest news. Instead, we engaged in heated discussions about politics, philosophy, and the meaning of life.

THE ANALOG PLEASURES:

Hostel life without the Internet was a return to analog pleasures. We read dog-eared novels borrowed from fellow travelers, scribbled postcards to loved ones, and strummed chords on the guitar under the night sky.

THE UNPLUGGED CREATIVITY:

In the absence of distractions, our creativity flourished. We sketched in notebooks, composed poems inspired by rustling leaves, and wove stories that danced in the moonlight.

THE LONGING:

Yet, there were moments when the longing for the digital world crept in. We missed instant communication and familiar memes. But we carried with us the lessons learned during those hostel nights—a chance to connect with ourselves and others in ways we had forgotten.

And so, we stepped out into the world, our backpacks heavy with memories and our hearts lighter without the constant buzz of notifications. Hostel life without the Internet had transformed us, leaving an indelible mark on our souls—a reminder that sometimes, the most profound connections are forged in the quiet spaces between Wi-Fi signals.



Poems



ANTICIPATION

How beautiful the earth is still, To the-how full of
happiness! How little fraught with real ill, Or unreal
phantoms of distress! How spring can bring the
glory, yet, And summer win the to forget
December's sullen time!

Why dost though hold the treasure fast, Of youth's
delight, when youth is past, And though art near
they prime?

When those who were they own compeers, Equals in
fortune and in years, Have seen their morning melt
in tears, To clouded, smileless day; Blest, had they
died untried and young, Before their hearts went
wandering wrong, - Poor slaves, subdued by
passions strong, A weak and helpless pray!

Because, I hoped while they enjoyed, And by
fulfilment, hope destroyed; As children hope, with
trustful breast, I waited bliss-and cherished rest. A
thoughtful spirit taught me soon, That we must long
till life be done;

ప్రయాణం

మరో ప్రపంచం, మరో ప్రపంచం, మరో ప్రపంచం పిలిచింది!
పదండి ముందుకు, పదండి త్రోసుకు!
పోదాం, పోదాం పైపైకి!

దారి పొడవునా గుండె నెత్తురులు తర్పణ చేస్తూ
పదండి ముందుకు
బాటలు నడచి పేటలు కడచి
కోటలన్నిటిని దాటండి.

నదీనదాలు, అడవులు, కొండలు
ఎడారులా మనకడ్డంకి?
ప్రభంజనం వలె హోరెత్తండి
భావవేగమున ప్రసరించండి.

త్రాచులవలెనూ రేచులవలెనూ
ధనుంజయునిలా సాగండి.
మరో ప్రపంచం మరో ప్రపంచం
మరో ప్రపంచం పిలిచింది.

పదండి ముందుకు పడండి త్రోసుకు
పోదాం పోదాం పైపైకి!

அம்மா

அம்மா என்று அழைக்கையிலே
அன்பின் வெள்ளம் பெருகுதடி
உன்னைக் கண்டு உலகம் மறந்து
உள்ளம் களிப்பு கொள்ளுதடி

கைகள் தாங்கும் கனிவுடனே
காலம் எல்லாம் கடந்து வரும்
முகம் காணும் பொழுதினிலே
முத்தம் கொண்டு மகிழ்விக்கும்

உன்னை நோக்கி உயிர் வாழ்க்கை
உன்னால் தானே உலகம் தொடும்
உன்னைச் சூழ்ந்து உறவுகள் எல்லாம்
உன்னால் தானே உயிர் பெறும்

குழந்தை வளரும் கூட்டிலேயே
குறைவு தீர்த்து குளிர் செய்து
கைகள் பட்டு கனிவு செய்து
காதல் வளர்த்து வளர்க்கும் தாய்

உன்னைக் காணும் பொழுதெல்லாம்
உள்ளம் பூரிப்பு அடைகின்றது
உன்னைக் கொண்டு உலகை அறிந்து
உள்ளம் விரிந்து பரவுகின்றது

அம்மா என்ற அழகிய சொல்
அன்பின் முழுமை தருகின்றது
உன்னைக் கொண்டு உலகை வெல்லும்
உள்ளம் கொண்ட மகன் நான்



Story



THE BROKEN BRIDGE

In a quaint village nestled between rolling hills, there existed an old stone bridge—the Willowbrook Bridge. For generations, it had connected the two sides of the village, allowing farmers to transport their produce and children to reach the school on time.

The bridge was more than just a structure; it was a symbol of unity. Its moss-covered stones whispered tales of love, laughter, and shared burdens. But as the years passed, the bridge began to crumble. The villagers patched it up, but the cracks widened, and the woodwork sagged.

One stormy night, a young woman named Lena stood at the bridge's edge. She had grown up hearing stories about its magic—the way it brought people together, the promises made under its arches. But now, she saw its decay—the missing planks, the frayed ropes, the danger lurking below.

Lena's heart ached. She had dreams beyond the village—college, adventure, a life beyond the hills. But her aging parents needed her. They ran the village bakery, and without Lena, it would crumble like the bridge.

One morning, as Lena kneaded dough, an idea struck her. She approached the village council, her eyes determined. "We must repair the Willowbrook Bridge," she declared. "It's our lifeline."

The council members exchanged glances. "Lena," said the elder, Mr. Harrington, "we lack the funds. The bridge is beyond repair."

But Lena persisted. She organized bake sales, talent shows, and community picnics. The villagers rallied, and slowly, the coffers filled. They hired skilled craftsmen, and day by day, the bridge transformed. New planks replaced the old, and fresh ropes held it together.



As the bridge took shape, something magical happened. People crossed it with lighter steps. Children played tag, lovers held hands, and elderly couples reminisced about their youth. The Willowbrook Bridge became more than a path—it became a testament to resilience and community spirit. One evening, Lena stood on the newly restored bridge, watching the sunset. Mr. Harrington joined her. “You’ve done something remarkable,” he said. “But why?”

Lena smiled. “Because bridges connect,” she replied. “Not just land, but hearts. We needed this bridge, not just for practical reasons, but to remind us that together, we can mend what’s broken.”

Mr. Harrington nodded. “And the cost?”

Lena’s gaze swept the horizon. “The cost,” she said, “was realizing that sometimes, sacrifices are bridges too. I gave up my dreams for this village, but in doing so, I found purpose.”

The villagers celebrated the bridge’s completion. They held a grand feast, and Lena danced with her parents under the moonlight. As the music played, she understood the true moral of her story:

“Sometimes, the most valuable bridges we build are not of stone or wood, but of love, compassion, and shared effort.”

And so, the Willowbrook Bridge stood—a testament to Lena’s sacrifice and the resilience of a community that refused to let anything crumble.



Thumbeti Srikanth
9922005158
II-ECE

THE CRACKED POT

Once upon a time there was a water-bearer in India who had two large pots, each hung on each end of a pole which he carried across his neck. One of the pots had a crack in it, and while the other pot was perfect and always delivered a full portion of water at the end of the long walk from the stream to the master's house, the cracked pot arrived only half full. For a full two years this went on daily, with the bearer delivering only one and a half pot full of water in his master's house.

Of course, the perfect pot was proud of its accomplishments, perfect to the end for which it was made. But the poor cracked pot was ashamed of its own imperfection, and miserable that it was able to accomplish only half of what it had been made to do.

After two years of what it perceived to be a bitter failure, it spoke to the water-bearer one day by the stream. "I am ashamed of myself, and I want to apologize to you." "Why?" asked the bearer. "What are you ashamed of?" "I have been able, for these past two years, to deliver only half my load because this crack in my side causes water to leak out all the way back to your master's house. Because of my flaws, you have to do all of this work and you don't get full value from your efforts, the pot said. The water-bearer felt sorry for the old cracked pot, and in his compassion he said, "As we return to the master's house, I want you to notice the beautiful flowers along the path."

Indeed, as they went up the hill, the old cracked pot took notice of the sun warming the beautiful wild flowers on the side of the path, and this cheered it some.



But at the end of the trail, it still felt bad because it had leaked out half its load, and so again it apologized to the bearer for its failure. The bearer said to the pot, "Did you notice that there were flowers only on your side of your path, but not on the other pot's side? That's because I have always known about your flaw, and I took advantage of it. I planted flower seeds on your side of the path, and every day while we walk back from the stream, you've watered them. For two years I have been able to pick these beautiful flowers to decorate my master's table. Without you being just the way you are, he would not have this beauty to grace his house."



Moral

Each of us has our own unique flaw. But it's the cracks and flaws we each have that make our lives together so very interesting and rewarding. You've just got to take each person for what they are and look for the good in them.

G. Sathya Deepak
9922005190
II-ECE-C

THE BOY AND THE APPLE TREE

Long time ago, there was a huge apple tree. A little boy loved to come and play around it everyday. He climbed to the treetop, ate the apples, and took a nap under the shadow. He loved the tree and the tree loved to play with him. Time went by, the little boy had grown up and he no longer played around the tree every day.

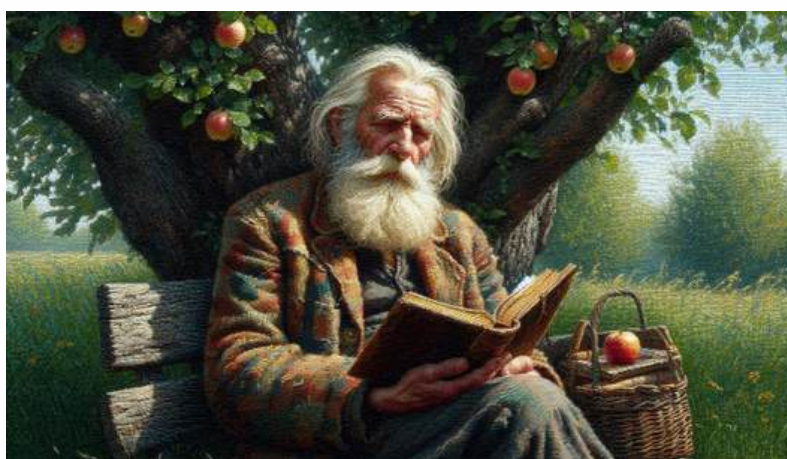


One day, the boy came back to the tree and he looked sad. "Come and play with me", the tree asked the boy. "I am no longer a kid, I do not play around trees any more" the boy replied. "I want toys. I need money to buy them." "Sorry, but I do not have money, but you can pick all my apples and sell them. So, you will have money." The boy was so excited. He grabbed all the apples on the tree and left happily. The boy never came back after he picked the apples. The tree was sad.

"Come and play with me" the tree said. "I do not have time to play. I have to work for my family. We need a house for shelter. Can you help me?" "Sorry, but I do not have any house. But you can chop off my branches to build your house." So the man cut all the branches of the tree and left happily. The tree was glad to see him happy but the man never came back since then. The tree was again lonely and sad. One hot summer day, the man returned and the tree was delighted. "Come and play with me!" the tree said. "I am getting old. I want to go sailing to relax myself. Can you give me a boat?" said the man. "Use my trunk to build your boat. You can sail far away and be happy." So the man cut the tree trunk to make a boat. He went sailing and never showed up for a long time. Finally, the man returned after many years.

"Sorry, my boy. But I do not have anything for you anymore. No more apples for you", the tree said. "No problem, I do not have any teeth to bite" the man replied. "No more trunk for you to climb on." "I am too old for that now" the man said. "I really cannot give you anything, the only thing left is my dying roots," the tree said with tears. "I do not need much now, just a place to rest. I am tired after all these years," the man replied.

"Good! Old tree roots are the best place to lean on and rest, come sit down with me and rest." The man sat down and the tree was glad and smiled with tears.



Moral

The story reflects how we often take our parents for granted, only relying on them when needed or in trouble. The tree symbolizes the cruel treatment we give our parents, who provide everything to make us happy until it's too late.

Devisetty Ruchitha

9922005

II-ECE-C

A large, white, circular graphic with a torn, deckle-like edge. The words "Pencil Art" are written in a black, elegant cursive script across the center of the circle. The background of the entire page is a photograph of an art studio. On the right, a wooden easel holds a canvas with a faint, abstract pencil sketch. To the left of the circle, a green monstera plant is visible. In the foreground, a dark, round wooden table holds a small jar of black ink, a pencil, and some rough, light-colored stones. The lighting is warm and focused on the central elements.

Pencil Art



Y Rama Krishna



S Veda Sri Varshitha



S. Sankarappan Tamizhenth





S. Sankarappan Tamizhenth



P Vamsi Krishna



S. Sankarappan Tamizhenth





P Vamsi Krishna



P Vamsi Krishna



Y Rama Krishna





Y Rama Krishna



P Vamsi Krishna



Y Rama Krishna





P Vamsi Krishna



Y Rama Krishna



P Vamsi Krishna





Photography



N Navaneeth - III C



Vijay Kumar - III E



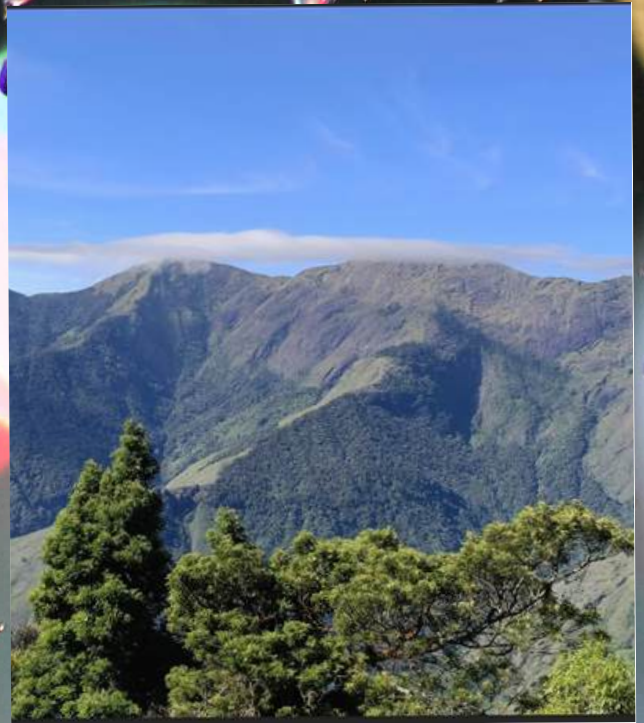
Giridhar - II C



N Naveenth - III C



M Jyoshitha - I ECE



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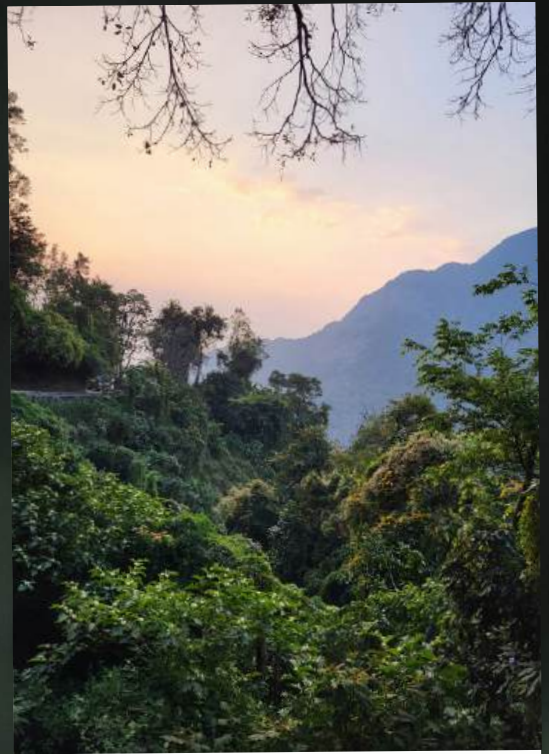
Saniya Reddy - I ECE



D Varun Sai - I ECE



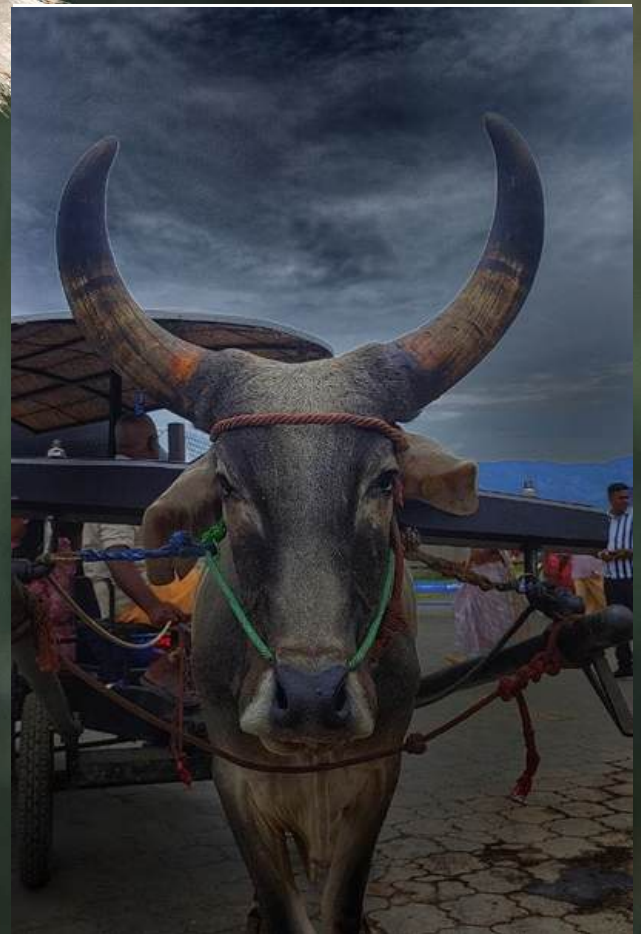
K Pavan Kumar - III A



S Teja II A



G Siva Prasad - III A



T Abhinav - III D

FEEDBACK FROM OUTGOING STUDENTS

I am delighted to provide my feedback regarding our Institution. I entered university as a zero-knowledge student with a phobia of connection and communication, but my four years here taught me about education with ethics. Me as an ECE student, had the best teachers who taught me the route to success. Life provides you with a book of maps, but picking the right way can help you get to your destination, as my lecturers explained. It's a learning adventure to become the best version of myself, which can only be achieved with the help and encouragement of my department instructors. I am grateful for the opportunity to provide input to the college that has helped shape my career. Thankyou.



K. Rachana Chowdary
IV - ECE - C



K. Thrimurthy Reddy
IV - ECE - C

Our ECE department at our institution is distinguished by devoted faculty members who demonstrate exceptional teaching skills and provide strong assistance to students. They provide valuable hands-on learning experiences through lab sessions and projects, fostering an environment that promotes exploration and growth. Furthermore, their work with Nanochip has been quite beneficial, allowing students to explore deeper into their hobbies. I'd want to convey my heartfelt gratitude to our ECE department and institution for their persistent devotion to our education. Their drive to excellence has helped shape our academic path and prepared us for future ventures.

The ECE department has been a pleasant experience, with kind and knowledgeable lecturers offering help and project ideas. The curriculum provides a strong basis in a variety of topics, but real-world applications and practical experience are more vital. Students benefit from industry ties and internships, which help prepare them for future employment. Attending department events, as well as technical and cultural events, improves the whole experience. Overall, the student is pleased with their education and feels prepared for future activities.



Greeshma Malasani
IV - ECE - B

I am overjoyed to have earned my degree in the ECE department at KARE, and I am also grateful for the training from industry experts who have collaborated with the department. I am grateful to my faculty members for their continued support and guidance, which has helped me establish a successful profession.

I'd like to convey my heartfelt gratitude to the Kalasalingam University administration for providing me with a joyful campus environment...



Samara Simha Reddy
IV - ECE - D

The department's extensive curriculum, friendly professors, and emphasis on research possibilities, industry partnership, and community engagement have all enhanced my academic path, offering valuable skills and experiences for future ventures. Faculty members are committed and supportive. I am appreciative for the assistance and direction I received during my stay in the department. The department's commitment to excellence and innovation creates an ideal learning environment for students to succeed in the dynamic field of electronics and communication engineering.



Rabiya
IV - ECE - D

STUDENT ACHEIVEMENTS



Two of Our students Nambi E and Gowshigan S were honored with an Outstanding Research Award in B.Tech (Electronics and Communications Engineering) for the years 2020-2024



M. Sai Ganesh from III year ECE, has got score card in GATE 2024



Bhavesb Balaji M, Guhanesh Kumar, Hemanth Reddy, Avulapati Giridhar had received Best NSS volunteers Awards for Dept. of ECE



Third year Students have won the First Prize in Nanothon 2.0 conducted by Nanochip solutions

Placements

Our Department has an abundance of job opportunities available to students in both core and software industries, with several renowned companies like genpact, neyes, zuci systems, comcast, accenture, zifo, foxconn, tcs and more are being potential employers.

OUR RECRUITER'S



Placements



genpact

-11 LPA



PULI PAVAN KUMAR
IV - ECE - C



MUPPARAJU HEMA PRIYA
IV - ECE - C



COMCAST

-5 TO 8LPA



**ALAVALAPATI
PRASHANTH REDDY**
IV - ECE - C



**GOWSHIGAN
SOKKALINGAM**
IV - ECE - A

GALLERY

Our department regularly organizes workshops, events, and celebrations for a well-rounded academic experience. We are proud of our students' success in external college events and believe these provide valuable real-world opportunities for growth.

EXTRAMURAL EVENTS

MATLAB MAINA



A competition in which both external and internal students demonstrated their great MATLAB talents.

ELECTROMAKER



An event was held in which internal and external students competed to present their projects relevant to the autonomous car sector.

IOTRONZ



ELECSHPERE ODYSSEY



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COCURRICULAR EVENTS

SOCCER ROBOT WORKSHOP



A Workshop was conducted with the collaboration of Aadhya Skills , where students learned about the National Instruments tools like MyRio and Labview.

ISRO WORKSHOP



A Workshop was conducted with the collaboration of ISRO, where students learned about the Importance Artificial Intelligence in Space Applications

ONE CREDIT COURSE



A glance of One Credit Course on Advanced Machine Learning for Underwater Image Processing offered by Our Department

EXTRACURRICULAR EVENTS

PONGAL CELEBRATION



WOMEN'S DAY CELEBRATION





"I want to tell the youth, don't just focus on your future but also lay the foundation of the country's future. As President, you have my full support."

- Droupadi Murmu